



By Bhavani:

1. What we need to start Floor plan?

Ans: To start a floor plan first we need inputs like .v, .lib, .lef, .SDC

This is the first major step in getting your layout done. Your floor plan determines your chip quality. At this step, you define the size of your chip/block, allocates power routing resources, place the hard macros, and reserve space for standard cells

2. Tell about input files?

Ans: NETLIST:

It is the combination of sequential elements and their logical connectivity.

Netlist contains-

- Input and output information of the design.
- Wire information.
- Cell and instance information.
- Module information.
- Hierarchy information.
- Port information.

LIBRARY file:

It has time, power and functionality of a cell

Time- input delays, output delays, setup and hold

Power- leakage power and internal power

Cell functionality

For eg:- $(A+B)*(B+C)$

PVT(process voltage and temperature) conditions

LEF(library exchange format)

It has physical information of the design

Two types

1. Technology lef
 2. Cell/macro lef
1. Technology lef
- It contains metal layer and via information like
Metal layer:

- Direction
- Pitch
- Width
- Area
- Spacing table
- Min enclosure area
- Diag spacing
- Diag min edge length
- Resistance
- Capacitance
- Thickness
- Antenna model and antenna area ratio
- DC current density

VIA information:



- Spacing
- Width
- Antenna model
- Antenna area ratio
- DC current Density

2. Cell/Macro lef

- Class
- Origin
- Size
- Symmetry
- Pin:
 - Antenna gate area
 - Direction
 - Usage
 - Port

SDC (Synopsys design constraint)

- Clock definition
 - Create clock
 - Create virtual clock
 - Create generated clock
 - Create clock uncertainty
- External delays
 - Input delays
 - Output delays
- DRV's

Max tran, max cap and max fanout

- Timing path exceptions
 - False path
 - Multi cycle path
 - Max delay
 - Min delay

3. What are the guidelines to place macros?

Ans: 1. Place macros around chip periphery.

If you don't have reasonable rationale to place the macro inside the core area, then place macros around the chip periphery. Placing a macro inside the core can invite serious consequence during routing due to a lot of detour routing, because macros are equal to a large obstacle for routing. Another advantage to placing the hard macros around the core periphery is it's easier to supply power to them, and reduces the change of IR drop problems to macros consuming high amounts of power.

2. Consider connections to fixed cells when placing macros.

When you decide macro position, you have to pay attention to connections to fixed elements such as I/O and perplaced macros. Place macros near their associate fixed element. Check connections by displaying flight lines in the GUI.

3. Orient macros to minimize distance between pins.

When you decide the orientation of macros, you also have to take account of pins positions and their connections.

4. Reserve enough room around macros.

For regular net routing and power grid, you have to reserve enough routing space around macros. In this case estimating routing resources with precision is very important. Use the congestion map from trialRoute to identify hot spots between macros and adjust their placement as needed.

5. Reduce open fields as much as possible.

Except for reserved routing resources, remove dead space to increase the area for random logic. Choosing different aspect ratio (if that option is available) can eliminate open fields.

6. Reserve space for power grid.

The number of power routes required can change based on power consumption. You have to estimate the power consumption and reserve enough room for the power grid. If you underestimate the space required for power routing, you can encounter routing problems.

4. what happens if pins assign to left and right.(if you have IO pins at top and bottom)?

Ans: Actually top level chip will be divided into some blocks, IO pins will be placed according to the communication between surrounding blocks.

If we assign pins to left and right rather than top and bottom we will face routing issues in further stages.

5. How we will assign spacing between two macros?

Ans: $\text{channel spacing} = \frac{\text{no of pins} \times \text{pitch}}{\text{total number of metal layers}}$

6. In placement what are the congestion types, how to resolve congestion?

Ans: we will see congestion where available tracks are less than required tracks.

We may see congestion because of

- Cell density
- Pin density
- Bad floorplan

There are two types of congestion

1. horizontal congestion and
2. vertical congestion

we will see horizontal congestion when horizontal tracks are less and similarly for vertical congestion if vertical tracks are less

prevention techniques:

- we can avoid congestion by placing blockages.
- Cell padding
- Scan chain reordering.

7. What happens if cell density and pin density is more, how to resolve it?

Ans: if cell density and pin density is more we will see congestion and routing issues.

By placing partial blockage we can avoid cell density and by cell padding we can avoid pin density.

8. what happens if cells place closer to macros?

Ans: if cells are placed close to macros we will see routing issues near macros, to avoid this we are placing Halo around the macro.

9. explain about power planning?

Ans: Power Planning is one of the most important stage in Physical design. Power network is being synthesized, It is used provide power to macros and standard cells within the given IR-Drop limit. Steady state IR Drop is caused by the resistance of the metal wires comprising the power distribution network. By reducing the voltage difference between local power and ground, steady-state IR Drop reduces both the speed and noise immunity of the local cells and macros.

Power planning management can be divided in two major category first one is core cell power management and second one I/O cell power management. In core cell power planning power rings are formed around the core and macro. In IO cell power planning power rings are formed for I/O cells and trunks are created between core power ring and power pads.

In addition trunks are also created for macros as per the power requirement.

power planning is part of floor plan stage. In power plan, offset value for rings around the core and vertical and horizontal straps is being define

I/O cell library contains I/O cell and VDD/VSS pad cell libraries. It also contain IP libraries for reusable IP like RAM, ROM and other pre designed, standard, complex blocks.

10. what happens if IO pins placed at core boundary?

Ans: nothing happen, we can place IO pins in core boundary

11. what are the timing issues after placement?

Ans: DRV's and setup.

12. what is path timing and data path?

Ans: Timing Path

Timing path is defined as the path between start point and end point where start point and end point is defined as follows:

Start Point:

All input ports or clock pins of a sequential element are considered as valid start point.

End Point:

All output port or D pin of sequential element is considered as End point

For STA design is split into different timing path and each timing path delay is calculated based on gate delays and net delays. In timing path data gets launched and traverses through combinational elements and stops when it encounter a sequential element. In any timing path, in general (there are exceptions); delay requirements should be satisfied within a clock cycle.

In a timing path wherein start point is sequential element and end point is sequential element, if these two sequential elements are triggered by two different clocks(i.e. asynchronous) then a common least common multiple (LCM) of these two different clock periods should be considered to find the launch edge and capture edge for setup and hold timing analysis.

Different Timing Paths

Any synchronous design is split into various timing paths and each timing path is verified for its timing requirements. In general four types of timing paths can be identified in a synchronous design. They are:

Input to Register

Input to Output

Register to Register

Register to Output

Data path:

The path wherein data traverses is known as data path. Data path is a pure combinational path. It can have any basic combinational gates or group of gates.

13. Explain about PD flow?

Ans: INPUTS:

.v, .lib, .lef, .SDC

NETLIST:

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LEF(library exchange format)

It has physical information of the design

Two types

3. Technology lef
4. Cell/macro lef
3. Technology lef

It contains metal layer and via information like

Metal layer:

- Direction
- Pitch
- Width
- Area
- Spacing table
- Min enclosure area
- Diag spacing
- Diag min edge length
- Resistance
- Capacitance
- Thickness
- Antenna model and antenna area ratio
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- Class
- Origin
- Size
- Symmetry
- Pin:
 - Antenna gate area
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Input delays
Output delays

- DRV's
 - Max tran, max cap and max fanout
- Timing path exceptions
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 - Max delay
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SANITY CHECKS:

1. Library checks
 - Missing cell information
 - Missing pin information
 - Duplicate cells
2. Design checks
 - Inputs with floating pins
 - Nets with tri-state drivers
 - Nets with multiple drivers
 - Combinational loops
 - Empty modules
 - Assign statements
3. Constraint checks
 - All flops are clocked or not
 - There should not be unconstraint paths
 - Input and output delays

FLOORPLAN:

1. Utilization factor decides the size of the block.
2. Aspect ratio gives shape of the block.
3. After utilization and aspect ratio we go for pin placement.
 - In pin placement we have to place pins legally
4. Macros should be placed according to guidelines
 - a. Place macros around chip periphery.

If you don't have reasonable rationale to place the macro inside the core area, then place macros around the chip periphery. Placing a macro inside the core can invite serious consequence during routing due to a lot of detour routing, because macros are equal to a large obstacle for routing. Another advantage to placing the hard macros around the core periphery is it's easier to supply power to them, and reduces the change of IR drop problems to macros consuming high amounts of power.
 - b. Consider connections to fixed cells when placing macros.

When you decide macro position, you have to pay attention to connections to fixed elements such as I/O and preplaced macros. Place macros near their associate fixed element. Check connections by displaying flight lines in the GUI.
 - c. Orient macros to minimize distance between pins.

When you decide the orientation of macros, you also have to take account of pins positions and their connections.



d. Reserve enough room around macros.

For regular net routing and power grid, you have to reserve enough routing space around macros. In this case estimating routing resources with precision is very important. Use the congestion map from trialRoute to identify hot spots between macros and adjust their placement as needed.

e. Reduce open fields as much as possible.

Except for reserved routing resources, remove dead space to increase the area for random logic. Choosing different aspect ratio (if that option is available) can eliminate open fields.

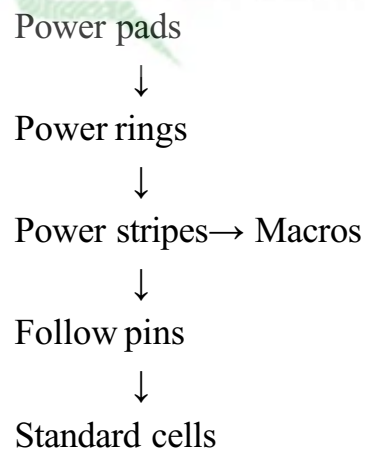
f. Reserve space for power grid.

The number of power routes required can change based on power consumption. You have to estimate the power consumption and reserve enough room for the power grid. If you underestimate the space required for power routing, you can encounter routing problems

5. After macro placement we will place physical cells like endcap and welltap cells

POWER PLANNING:

Power planning is to supply power to the standard cells and macros.



PLACEMENT:

Two stages- 1. Course placement 2. Detail placement

1. Course placement:

a. First tool will place standard cells based on hierarchy

b. It will do High fanout net synthesis

Adding buffers to the high fanouts

c. Scan chain reordering

In a less complex design, you don't usually do scan reordering. However, sometimes it may become difficult to pass scan timing constraints once the placement is done. The scan flip flop placements may create lengthier routes if the consecutive flops in scan chain are placed far apart due to a functional requirement. In this case, the PnR tool can reconnect the scan chains, to make routing easier. A prerequisite for this option is a scan DEF for the tool to recognize the chains.

d. Logical optimization

Sizing

VT swapping

Buffering



- Logic restructuring
- Pin swapping
- Cloning
- Rebuffering
- Trail route

2. Detail placement
 - a. Area recovery
 - b. Congestion driven
 - c. Time driven

PLACEMENT OPTIMIZATION:

In optimization tool will optimize DRV's and setup timing

Here we will not see hold because clock is ideal.

Checks in placement:

Cells legalization

Utilization

Area

Timing

Congestion



CLOCK TREE SYNTHESIS:

Before CTS we need to check:

1. All cells should be legalized.
2. All power nets are prerouted.
3. All pins should be legalized.
4. Congestion, timing should be controlled.

GOALS OF CTS:

1. To minimize the logical DRCs.
2. Balancing the skew.
3. Minimum Insertion Delay.

INPUTS OF CTS:

1. SDC
2. SPEC FILE
3. PLACEMENT DATABASE

WHAT IS CTS?

To distribute a clock from Clock port to Clock pin

WHY CTS?

To minimize skew and insertion delay to build the clock tree.

Here we are generating SPEC file using clockbuffers and clock inverters.

SPEC file consists of

1. Buffers list
2. Max skew
3. Min and Max Insertion delay
4. Max trans, Cap, Fanout
5. Inverters list
6. Clock tree leaf pin, exclude pin, stop pin
7. Clock name
8. Clock period

COMMANDS USED IN CTS:

clockdesign

optDesign -postCTS

CHECKS IN CTS:

1. Timing numbers
2. Utilization numbers
3. Congestion
4. All cells should legalize

ROUTING:

INPUTS:

CTS database

Captables

GOAL:

We need to interconnect all the nets without leaving shots and Spacing violations.

STEPS INVOLVED IN ROUTING:

1. Global Routing
2. Track Assignment
3. Detailed Routing

GLOBALROUTING:

Router breaks the routing portion of the design into rectangles called gcells and assigns signalnets to gcells.

The global router attempts to find shortest path through gcells but does not make actual connection or assign nets to specific nets and to specific track within gcell.

TRACKASSIGNMENT:

In this step the nets are properly assigned on tracks.

DETAILED ROUTING:

Nanoroute follows global routing plan and lays down actual wires that connect pins to their corresponding nets.

It creates shorts and opens or spacing violations rather than leaving unconnected nets.

We can route detailed routing on entire design, a specified area of design on selected nets.

Router runs SEARCH AND REPAIR ROUTING during detail routing.

It locates shorts and opens and spacing violations so, it reroutes the effected area to eliminate violations.

CHECKS:

1. Verify connectivity
2. Verify geometry
3. timing numbers
4. utilization numbers
5. All cells should legalize
6. Congestion

Commands:

Routedesign
optDesign -postRoute

14. how will you place macros according to hierarchy?

Ans: According to hierarchy communicating macros will be in same color, based on that we can place macros .

15. if we do macro abutment, what happens?

Ans: There are two cases

1. If two macros communicating only with each other we can abutment the macros
2. If the macros communicating with other cells(std cells and IO ports) then we must should provide a proper channel spacing between the macros or else we can see the routing issue

16. Can we place macros 90 and 270dergees orientation?

Ans: It depends on which technology you are working on. 45nm & below there are orientation requirements by foundry. Poly orientation should be same throughout the chip. So Macro poly orientation should match with the poly orientation of the standard cells.

17. In power planning for rings and stripes which metal layers used and why?

Ans: For rings and stripes we use top metal layers because for top metal layers we have low resistivity.

18. Can we place cells between the space of IO and core boundary?

Ans: No, we cannot place cells between the space of IO and core boundary because in between IO and core boundary power rings will be placed and we may see routing issues.

19. How did you placed standard cells with command and tool?

Ans: command: placeDesign

Tool: place → place standard cells

20. what type of congestion you seen after placement?

- Ans:
1. Congestion near Macro corners due to insufficient placement blockage.
 2. Standard cell placement in narrow channels led to congestion.
 3. Macros of same partition which are placed far apart can cause timing violation.
 4. Macro placement or macro channels is not proper.
 5. Placement blockages not given
 6. No Macro to Macro channel space given.
 7. High cell density
 8. High local utilization
 9. High number of complex cells like AOI/OAI cells which has more pin count are placed together.
 10. Placement of std cells near macros
 11. Logic optimization is not properly done.
 12. Pin density is more on edge of block
 13. Buffers added too many while optimization
 14. IO ports are crisscrossed; it needs to be properly aligned in order.

21. what are the physical cells?

Ans: **End Cap cells:**

1. These cells prevent the cell damage during fabrication.
2. Used for row connectivity and specifying row ending.
3. To avoid drain and source short.
4. These are used to address boundary N-Well issues for DRC cleanup.

Well Tap cells:

1. These are used to connect VDD and GND to substrate and N-Well respectively because it results in lesser drift to prevent latch-up.
2. If we keep well taps according to the specified distances, N-Well potential leads to proper electrical functioning.
3. To limit the resistance between power and ground connections to wells of the substrate.

De-cap Cells:

1. They are temporary capacitors which are added in the design between power and ground rails to counter the functional failure due to dynamic IR drop.

2. To avoid the flop which is far from the power source going into metastable state.

Filler Cells:

To fill the empty space and provide connectivity of N-wells and implant layers.

22. Tell about Non Default Rules?

Ans: Double width and double space.

After PNR stage if u will get timing /crosstalk/noise violations which are difficult to fix at ECO stage we can try this NDR option at routing stage.

USAGE OF NDRs and Example:

When we are routing special nets like clock we would like to provide more width and more spacing for them. Instead of default of 1unit spacing and 1unit width specified in tech file;But NDR having double spacing and double width .When clocknet is routed using NDR it has better Signal integrity, lesser crosstalk,lessnoise,but we cannot increase the spacing and width because it effects the area of the chip.

Double spacing: It is used to avoid the crosstalk.

Double width: It is used to avoid the EM.

23. What is setup and hold?

Ans: SETUP: Minimum time required for data stability before the clock edge.

HOLD: Minimum time required for data stability after the clock edge.

24. Can we do setup check at placement?

Ans: Yes, we will check setup in placement stage, where as we won't bother about hold because clock is idea in placement stage.

25. what is trail route and global route?

Ans: Trail route:

Trial Route performs quick global and detailed routing for estimating routing-related congestion and capacitance values. It also incorporates any changes made during placement, such as scan reorder. You can use Trial Route results to estimate and view routing congestion, and to estimate parasitic values for optimization and timing analysis. When used during prototyping, Trial Route creates actual wires, so you can get a good representation of RC and coupling for timing optimization at an early stage in the flow. Trial Route also produces a congestion map you can view to get early feedback on whether the design is routable. Trial Route results can also be used for pin assignment when you commit partitions.

Detaile route:

Detailed routing is where we specify the exact location of the wires/interconnects in the channels specified by the global routing. Metal Layer information of the interconnects are also specified here.

26. What is the cell height?

Ans: It is the height between two rows.

27. how to fix hold?

Ans: Hold fixing techniques:

- Downsizing
- VT swapping
- Pulling capture clock path
- Pushing launch clock path
- Insert buffer in data path

28. what is max tran range?

Ans: It is the range given in SDC file, if transition delay crosses that range we will see tran violations.

29. which technology is yours?

Ans: 45nm.



30. What is macro count, standard cell count and how many clocks in your design?

Ans: 4 macros, 36k standard cells and 3 clocks.

31. Already you placed macros, then you got core size X-10 and Y+10. How you place macros with command and from tool?

Ans: By command:

```
placeInst <macro name> {llx lly urx ury}
```

llx – Lower Left X co-ordinate

lly – Lower Left Y co-ordinate

urx – Upper Right X co-ordinate

ury – Upper Left Y co-ordinate

With Tool:

Go to Floorplan > Resize Floorplan and make the required changes to core and place the macros with the toolbar.

32. How to fix setup?

Ans: setup techniques:

- Downsizing
- VT swapping
- Pulling launch clock path
- Pushing capture clock path

33. Explain about isolation cells?

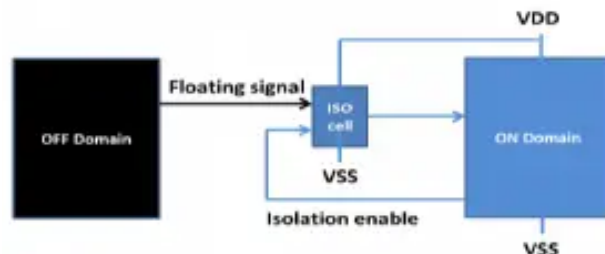
Ans: Isolation cells are used to isolate the output signals of a powered down domain. Output signals of a powered down domain has an intermediate voltage levels because of power gating effect. When such intermediate

voltage signals are feed as input to powered up domain, it could result in crowbar currents which affects the proper functioning of the powered up domain. Isolation cells helps to drive a valid logic value either zero or one.

Types

(1) Retention cells

(2) clamp cells



Clamp cells:

They are used to clamp the signals to a specified logic state.

- Clamp '0' type isolation cell is used to clamp the powered down output signal to the logic value of '0'. The circuit which can be used for this purpose can be something similar to a multiplexer. One input being the clamp value and other input being the signal to be isolated. Isolation Enable is the one which decides when to clamp the powered down signal hence it can be the select input to the multiplexer. The final optimal function which does this clamp '0' type Isolation is an AND gate with active low Isolation Enable.
- considering an active high isolation enable, an OR gate can be used as a clamp '1' Isolation cell. When the Isolation enable is high, the OR gate output is pulled to high irrespective of the other input signal. Again what makes it different from the normal OR gate is it is supplied with always ON supply or the power supply of the sink power domain.
- Retention cells: In majority of the cases the clamp value of the signal in the power down domain is determined by its RESET value. But there are some scenarios which warrant the clamp value to be same as the last logic state of the signal. This can be accomplished by using the retention type Isolation cells. A latch is required to store the last state value of the signal. Since the last state value can be either zero or one, the function required to implement the Isolation cell cannot be simplified further as we did for clamp 0 and clamp 1 type.

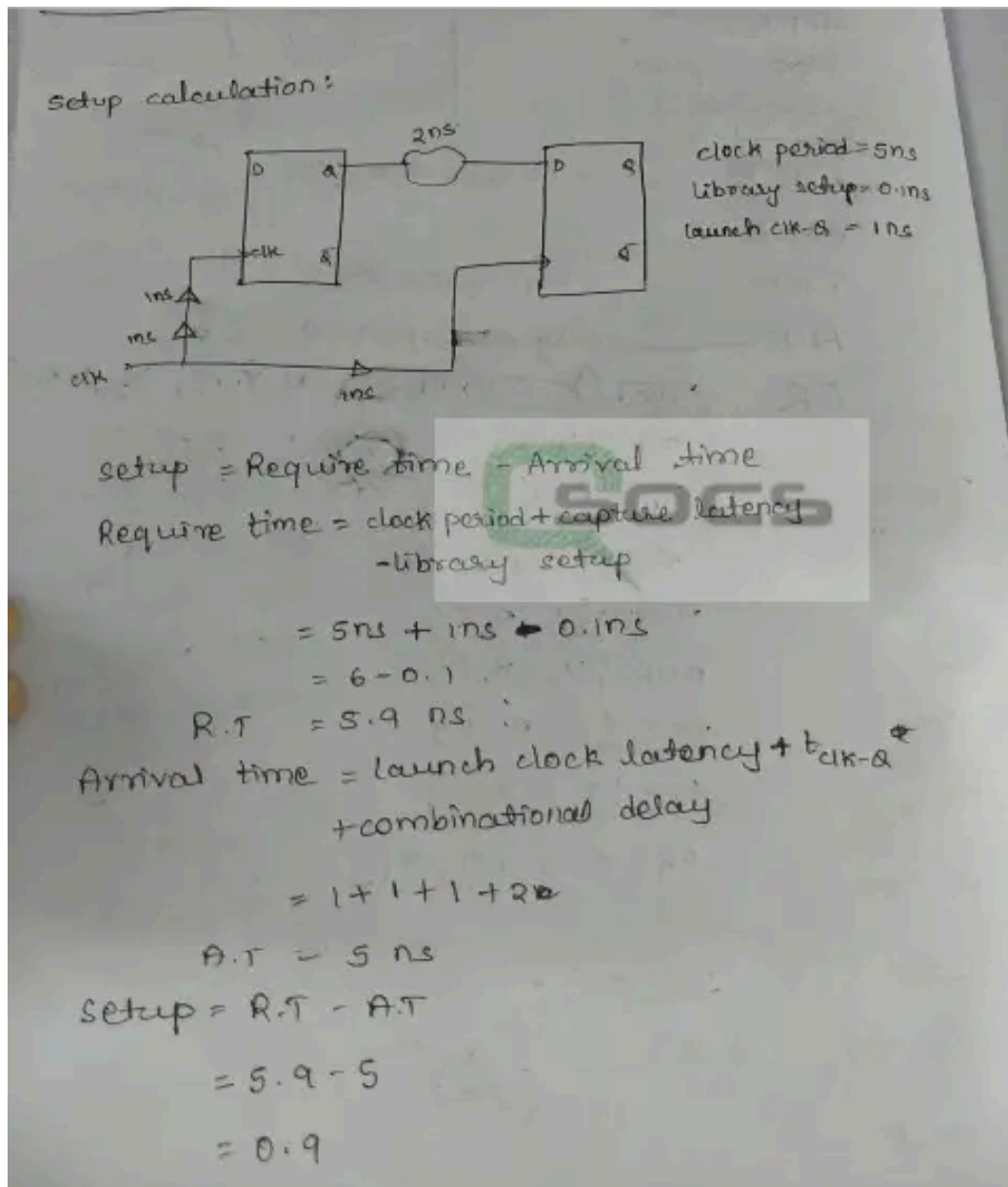
34. What is functional design and logical design?

Ans: Any chip designing can be subdivided in 2 steps

1. Front end or logic design- Based on specification provided, functionalities are created at RTL level abstraction to meet all the requirements. Generally Uses Gated logic. It is basically coding of digital design.

2. Back end or Physical design- After logic design and front end verification, in order to tape-out, the RTL abstraction is converted in form of transistors. They need to be optimised for low area, power and quality. This is physical or say analog design.

35. Setup calculation?

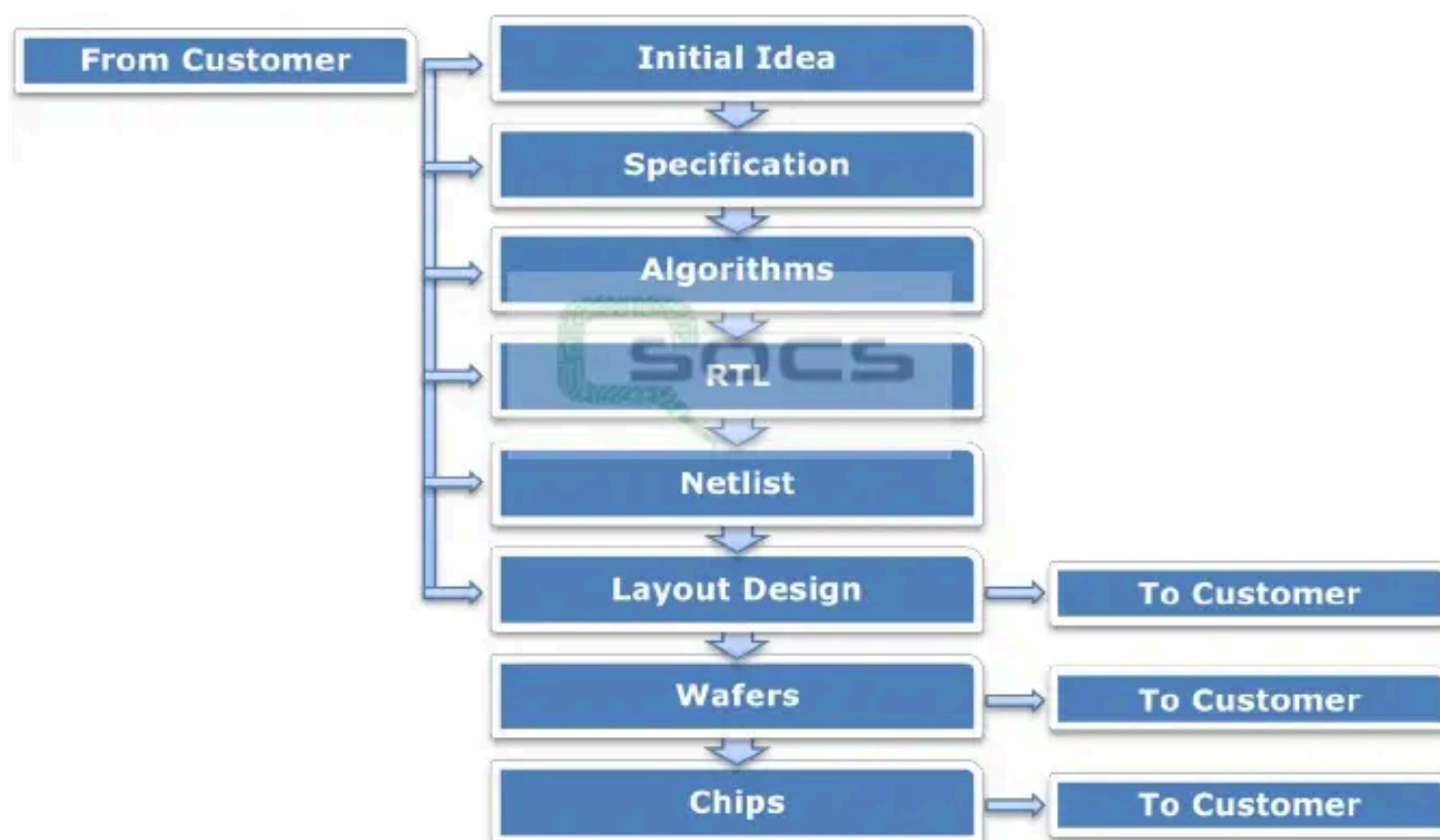


36. what are the inputs of all stages in PD flow?

- Ans: Floorplan: .v and .lef
- Placement: floorplan data base, lib.
- CTS: placement database, SDC and spec.
- Routing: CTS database, captables.

37. ASIC flow

Ans:



38. What spec file contains?

Ans: spec file contains:

- Clock name
- Clock period
- Max and min delay
- Max skew
- Sink max tran
- Buffer max tran
- Clock buffers and clock inverters information
- Exclude pin
- Through pin
- Information about Metal layers used

- Leaf route type

39. What is LVS?

Ans: Layout versus schematic

Inputs: .LVS.V, GDS II and rule deck file.

- Source netlist(physical) and reference netlist(logical) are converted in spice netlist
- LAYOUT: it takes source netlist into spice netlist
- SCHEMATIC: It takes reference netlist into spice netlist.
- LVS means comparison of layout and schematic spice netlist
- Spice netlist will count transistors and connectivity
- If layout and schematic netlist are equal we can proceed otherwise it will give below violations
 - Shorts
 - Opens
 - Floating nets
 - Pin mismatches
 - Component mismatch

40. Write setup and hold equations?

Ans: $\text{setup} = \text{require time} - \text{arrival time}$

Where $\text{require time} = \text{clock period} + \text{capture clock path latency} - \text{library setup} - \text{setup uncertainty}$

$\text{Arrival time} = \text{launch clock path latency} + \text{clock to Q delay} + \text{comb delay}$

$\text{Hold} = \text{arrival time} - \text{required time}$

Where $\text{require time} = \text{capture clock path latency} + \text{library hold} + \text{hold uncertainty}$

$\text{Arrival time} = \text{launch clock path latency} + \text{clock to Q delay} + \text{comb delay}$

41. How many master and generated clocks in your design?

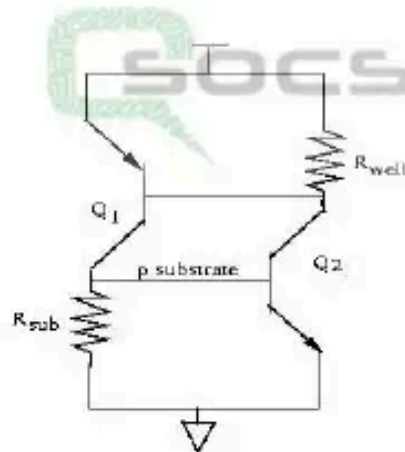
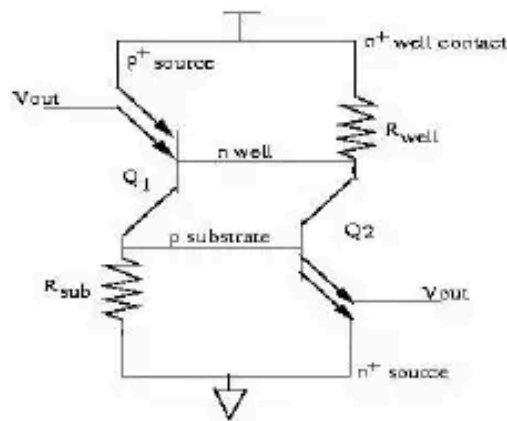
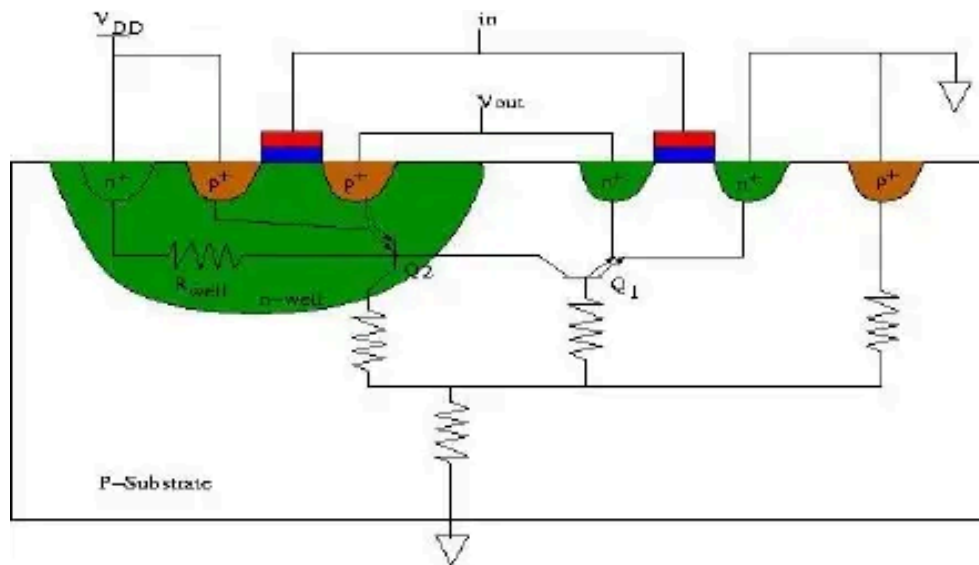
Ans: 2 master clocks and 1 generated clock

42. Explain latchup in CMOS?

Ans: Latch is the generation of a low-impedance path in CMOS chips between the power supply and the ground rails due to interaction of parasitic pnp and npn bipolar transistors. These BJT's form a silicon-controlled rectifier with positive feedback and virtually short circuit the power and the ground rail.

- This causes excessive current flows and potential permanent damage to the devices.

Analysis of the a CMOS Inverter CMOS depicting the parasitic



- The equivalent circuit shown has Q1 being a vertical double emitter pnp transistor whose base is formed by the n-well with a high base to collector current gain (β_1).
 - Q2 is a lateral double emitter npn transistor whose base is formed by the p-type substrate.
 - R_{well} represents the parasitic resistance in the n-well structure whose value ranges from 1KW to 20kW.
 - The substrate resistance R_{sub} depends on the substrate structure.
 - Assume the R_{well} and R_{sub} are significantly large so that they cause open circuit connections, this results in low current gains and the currents would be reverse leakage currents for both the npn and pnp transistors.
 - If some external disturbance occurs, causing the collector current of one of the parasitic transistors to increase, the resulting feedback loop causes the current perturbation to be multiplied by β_1, β_2
 - This event triggers the silicon-controlled rectifier and each transistor drives the other with positive feedback eventually creating and sustaining a low impedance path between power and the ground rails resulting in latch-up.
- For this condition if $\beta_1 * \beta_2$ is greater than or equal to 1 both transistors will continue to conduct saturation currents even after the triggering perturbation is no longer available.

- Some causes for latch-up are:
 - Slewing of V_{DD} during start-up causing enough displacement currents due to well junction capacitance in the substrate and well.
 - Large currents in the parasitic silicon-controlled rectifier in CMOS chips can occur when the input or output signal swings either far beyond the V_{DD} level or far below V_{SS} level, injecting a triggering current. Impedance mismatches in transmission lines can cause such disturbances in high speed circuits.
 - Electrostatic Discharge stress can cause latch-up by injecting minority carriers from the clamping device in the protection circuit into either the substrate or the well.
 - Sudden transient in power or ground buses may cause latch-up.

Guidelines For Avoiding Latch-Up

- Reduce the BJT gains by lowering the minority carrier lifetime through Gold doping of the substrate (solution might cause excessive leakage currents).
 - Use p^+ guard band rings connected to ground around nMOS transistors and n^+ guard rings connected to V_{DD} around pMOS transistors to reduce R_w and R_{sub} and to capture injected minority carriers before they reach the base of the parasitic BJT.
- Place substrate and well contacts as close as possible to the source connections of the MOS transistors to reduce the values of R_w and R_{sub} . (Solution to be used in your designs)
- Place source diffusion regions for the pMOS transistors so that they lie along equipotentials lines when currents flow between V_{DD} and p-wells.
 - Avoid forward biasing of the source/drain junctions so as not to inject high currents, this solution calls for the use of slightly doped epitaxial layer on top of the heavily doped substrate and has the effect of shunting the lateral currents from the vertical transistor through the low resistance substrate.

43. What are universal gates and why they are called as universal gates?

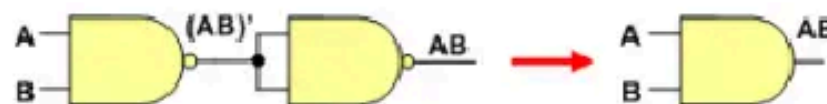
Ans: NOR gate and NAND gates have the particular property that any one of them can create any logical Boolean expression if designed in a proper way.

44. Implement AND gate with NAND gate?

Ans:

Implementing AND Using only NAND Gates

An AND gate can be replaced by NAND gates as shown in the figure (The AND is replaced by a NAND gate with its output complemented by a NAND gate inverter).



45. Full adder and its uses?

Ans: This type of adder is a little more difficult to implement than a half-adder. The main difference between a half-adder and a full-adder is that the full-adder has three inputs and two outputs. The first two inputs are A and B and the third input is an input carry designated as C_{IN} .

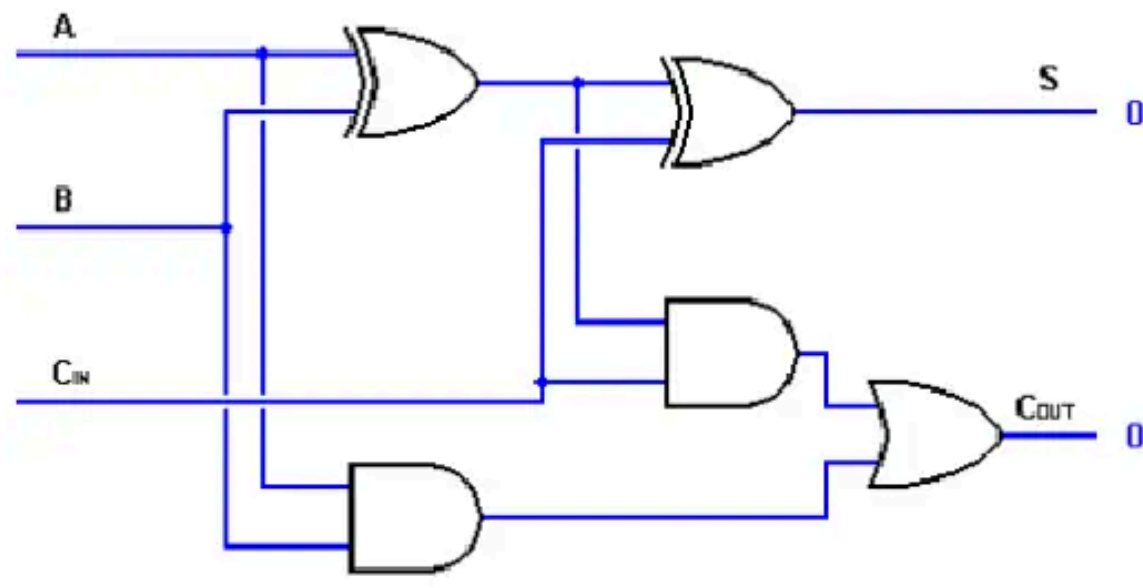
When a full adder logic is designed we will be able to string eight of them together to create a byte-wide adder and cascade the carry bit from one adder to the next.

The output carry is designated as COUT and the normal output is designated as S. Take a look at the truth-table.

INPUTS			OUTPUTS	
A	B	CIN	COUT	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

From the above truth-table, the full adder logic can be implemented. We can see that the output S is an EXOR between the input A and the half-adder SUM output with B and CIN inputs. We must also note that the COUT will only be true if any of the two inputs out of the three are HIGH.

Thus, we can implement a full adder circuit with the help of two half adder circuits. The first will half adder will be used to add A and B to produce a partial Sum. The second half adder logic can be used to add CIN to the Sum produced by the first half adder to get the final S output. If any of the half adder logic produces a carry, there will be an output carry. Thus, COUT will be an OR function of the half-adder Carry outputs. Take a look at the implementation of the full adder circuit shown below.



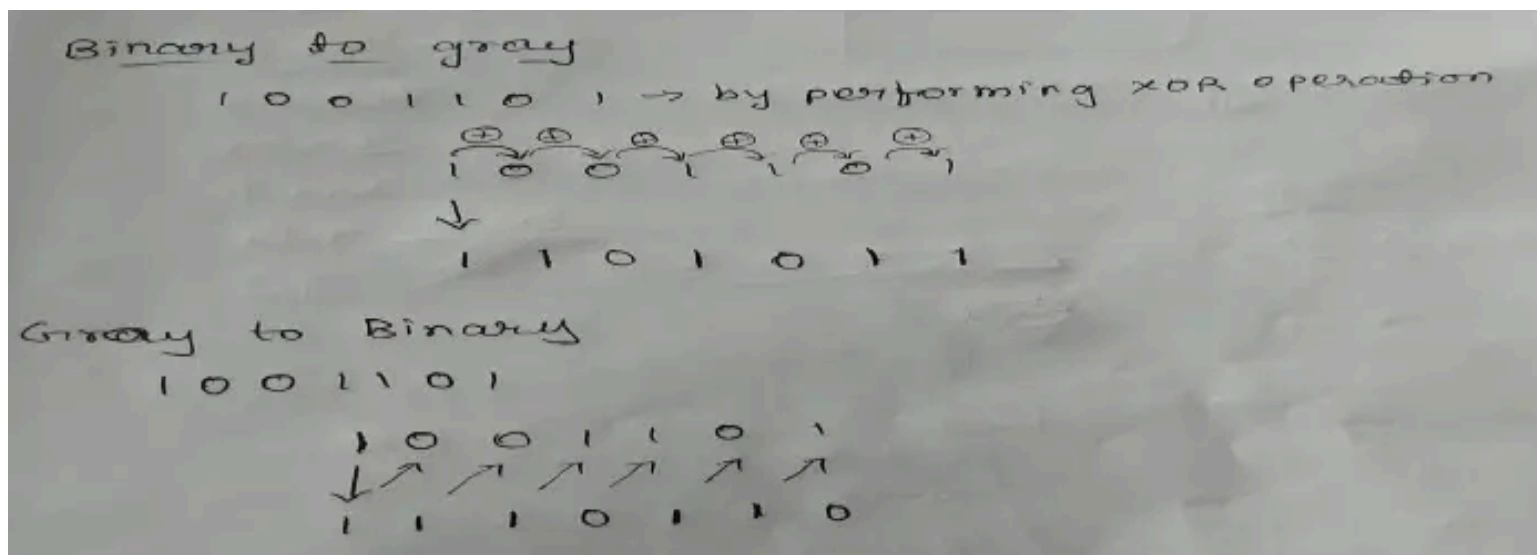
Uses of full adder:

Full adder reduces circuit complexity. It can be used to construct a ripple carry counter to add an n-bit number. Thus it is used in the ALU also. It is used in Processor chip like Snapdragon, Exynos or Intel pentium for CPU part. Which consists of ALU (Arithmetic Block unit). This Block is used to make operations like Add, subtract, Multiply etc. A full adder adds binary numbers and accounts for values carried in as well as out. A one-bit full adder adds three one-bit numbers, often written as A, B, and C_{in}; A and B are the operands, and C_{in} is a bit carried in from the previous less significant stage. The full adder is usually a component in a cascade of adders, which add 8, 16, 32, etc. bit binary numbers.

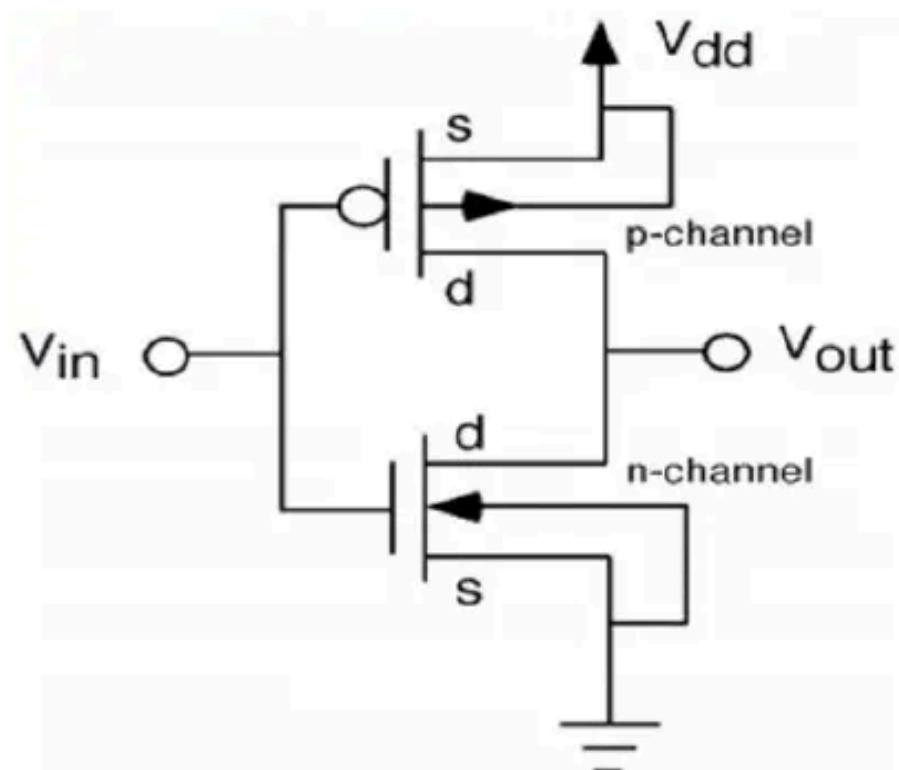
46. what are sequential and combinational circuits?

<u>Combinational Logic Circuits</u>	<u>Sequential Logic Circuits</u>
Output is a function of the present inputs (Time Independent Logic).	Output is a function of clock, present inputs and the previous states of the system.
Do not have the ability to store data (state).	Have memory to store the present states that is sent as control input (enable) for the next operation.
It does not require any feedback. It simply outputs the input according to the logic designed.	It involves feedback from output to input that is stored in the memory for the next operation.
Used mainly for Arithmetic and Boolean operations.	Used for storing data (and hence used in RAM).
Logic gates are the elementary building blocks.	Flip flops (binary storage device) are the elementary building unit.
Independent of clock and hence does not require triggering to operate.	Clocked (Triggered for operation with electronic pulses).
Example: Adder [1+0=1; Dependency only on present inputs i.e., 1 and 0].	Example: Counter [Previous O/P + 1=Current O/P; Dependency on present input as well as previous state].

47. binary to gray and gray to binary conversion?



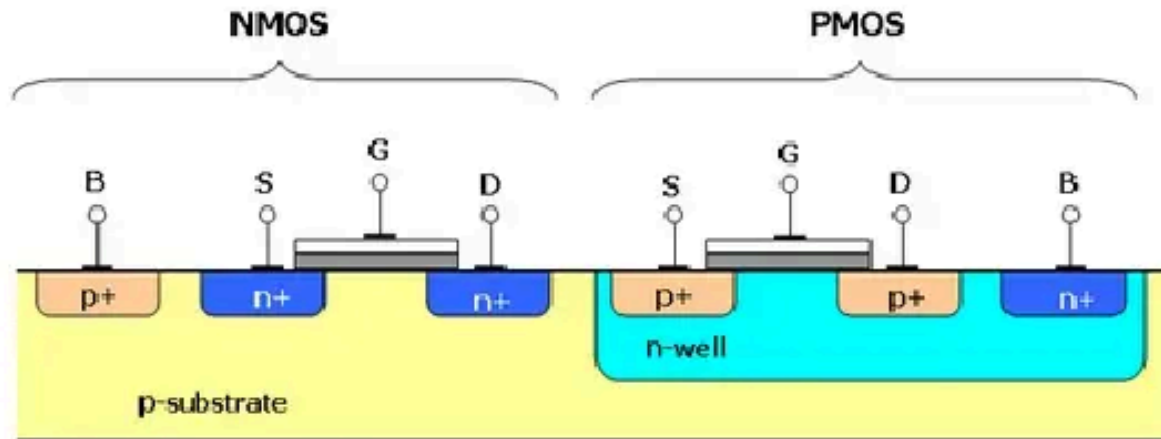
48. CMOS inverter



49. CMOS transistor explanation

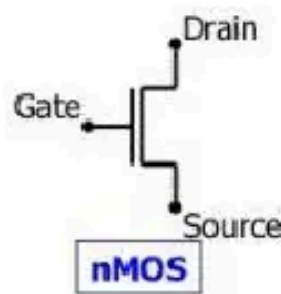
Ans: The main advantage of CMOS over NMOS and BIPOLAR technology is the much smaller power dissipation. Unlike NMOS or BIPOLAR circuits, a Complementary MOS circuit has almost no static power dissipation. Power is only dissipated in case the circuit actually switches. This allows integrating more CMOS gates on an IC than in NMOS or bipolar technology, resulting in much better performance.

Complementary Metal Oxide Semiconductor transistor consists P-channel MOS (PMOS) and N-channel MOS (NMOS).



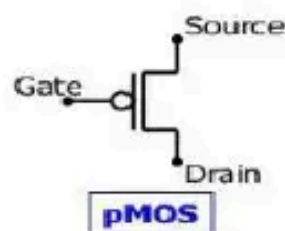
NMOS

NMOS is built on a p-type substrate with n-type source and drain diffused on it. In NMOS, the majority carriers are electrons. When a high voltage is applied to the gate, the NMOS will conduct. Similarly, when a low voltage is applied to the gate, NMOS will not conduct. NMOS are considered to be faster than PMOS, since the carriers in NMOS, which are electrons, travel twice as fast as the holes.



PMOS

P- channel MOSFET consists P-type Source and Drain diffused on an N-type substrate. Majority carriers are holes. When a high voltage is applied to the gate, the PMOS will not conduct. When a low voltage is applied to the gate, the PMOS will conduct. The PMOS devices are more immune to noise than NMOS devices.

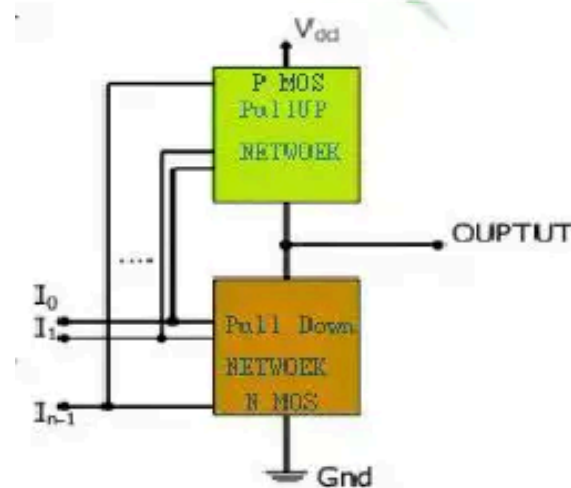


CMOS Working Principle

In CMOS technology, both N-type and P-type transistors are used to design logic functions. The same signal which turns ON a transistor of one type is used to turn OFF a transistor of the other type. This characteristic allows the design of logic devices using only simple switches, without the need for a pull-up resistor.

In CMOS logic gates a collection of n-type MOSFETs is arranged in a pull-down network between the output and the low voltage power supply rail (V_{ss} or quite often ground). Instead of the load resistor of NMOS logic gates, CMOS logic gates have a collection of p-type MOSFETs in a pull-up network between the output and the higher-voltage rail (often named V_{dd}).

Thus, if both a p-type and n-type transistor have their gates connected to the same input, the p-type MOSFET will be ON when the n-type MOSFET is OFF, and vice-versa. The networks are arranged such that one is ON and the other OFF for any input pattern as shown in the figure below.



CMOS offers relatively high speed, low power dissipation, high noise margins in both states, and will operate over a wide range of source and input voltages (provided the source voltage is fixed).

50. why we should not place macros in middle?

Ans: we will see more RC net delays and congestion.

51. In power plan, flip chip process?

Ans: Flip-chip is a method for interconnecting chips to external circuitry with solder bumps that have been deposited onto the chip pads. The solder bumps are deposited on the chip pads on the top side of the wafer during the final wafer processing step. In order to mount the chip to external circuitry (e.g., a circuit board or another chip or wafer), it is flipped over so that its top side faces down, and aligned so that its pads align with matching pads on the external circuit, and then the solder is flowed to complete the interconnect. This is in contrast to wire bonding, in which the chip is mounted upright and wires are used to interconnect the chip pads to external circuitry.

52. Why top layers for power, below layers for std. cells and middle layers for clock?

Ans: **Top metals layers:** The resistivity of top metal layers are less and hence less IR drop is seen in power distribution network. If power stripes are routed in lower metal layers this will use good amount of lower routing resources and therefore it can create routing congestion.

Middle metal layers: Middle routing layers such as 4,5 and 6 tend to have the same characteristics so the clock can be more predictable on those layers .Also ,fewer vias are required to connect to the metal to clock pin on the flop. It also require to metal layers but they are already reserved for power and GND.

Lower metal layers: Std cell require less power it will be available in lower metal layers. And some std cell are made up of lower metal layers. So no need to connect vias between std cell pins and metal layers.

53. What are the targets of placement?

- Ans:
- 1.Utilization
 2. Timing
 3. Congestion
 4. Area



54. Is SDC mandatory in floorplan?

Ans: no, because SDC contains clock definitions, delays, DRV's and exceptional paths so in floorplan we don't need all these information.

55. What information you see in SFEC and SPEF files?

Ans: SPEC contains:

- Clock name
- Clock period
- Max and min delay
- Max skew
- Sink max tran
- Buffer max tran
- Clock buffers and clock inverters information
- Exclude pin
- Through pin
- Information about Metal layers used
- Leaf route type

SPEF contains: RC values

56. What is the difference between normal buff & inverter and clock buff & clock inverter?

Ans: compare to normal buffers & inverters clock buffers & inverters have equal rise and fall time.

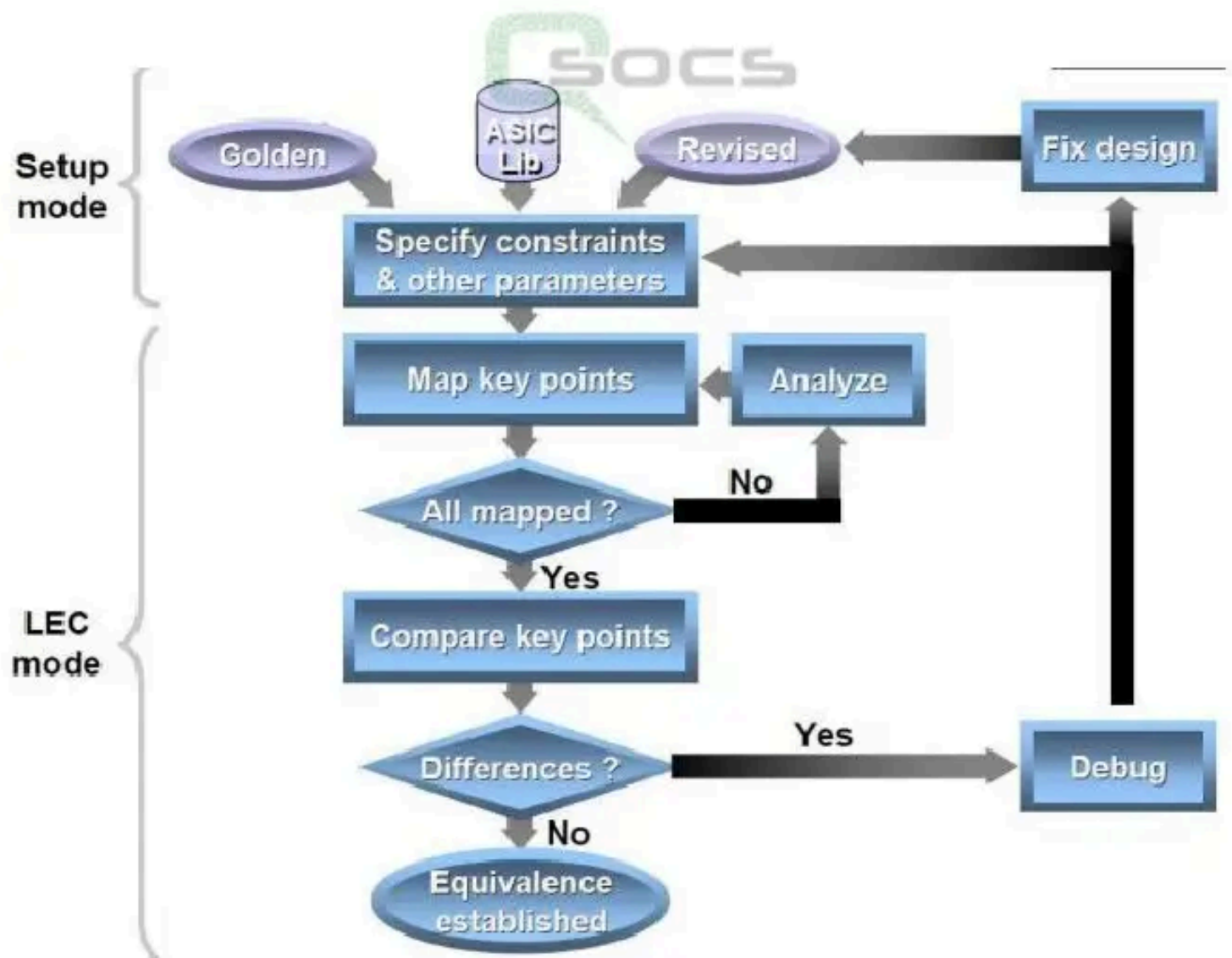
57. What are the outputs of powerplan?

Ans: power rings, power stripes, follow pins

We have to check DRC's

58. What is formal verification?

Ans:



59. Multi cycle path calculation?

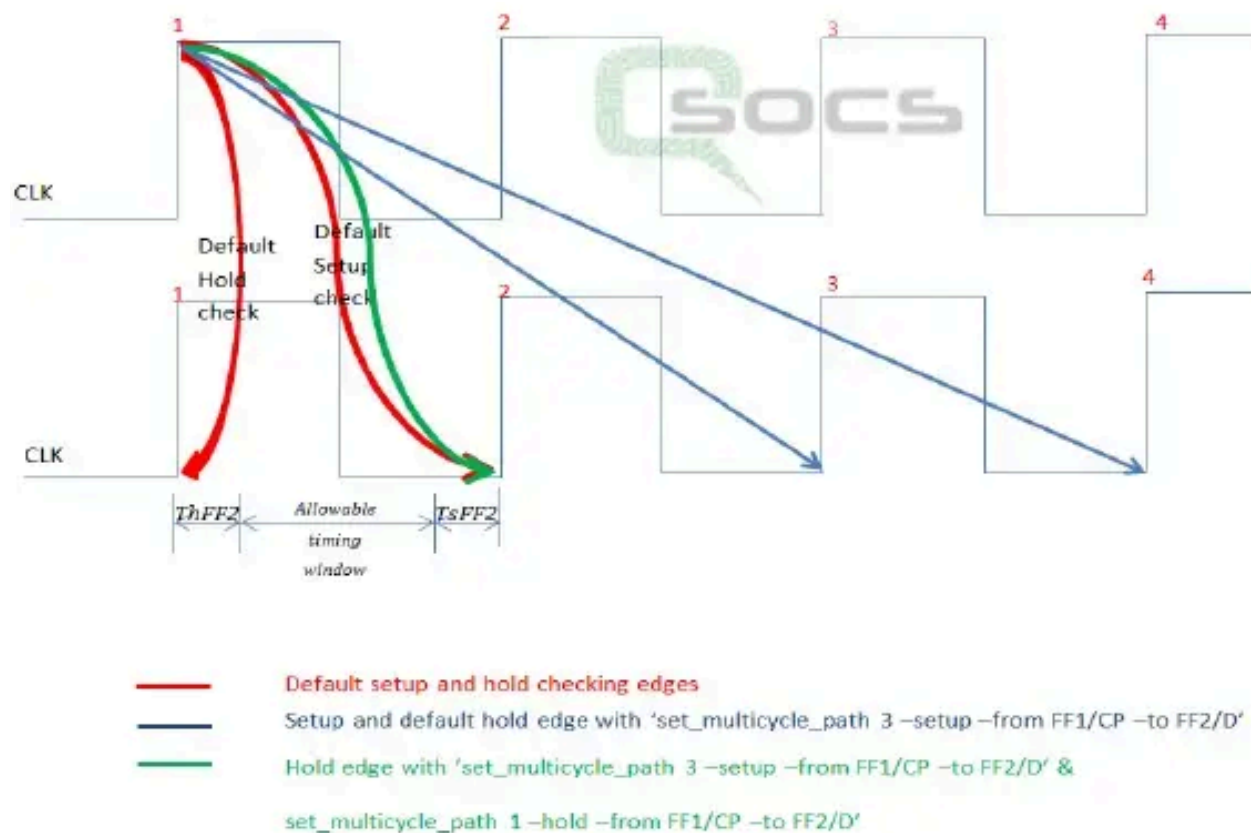


Ans: By default, we expect every timing path to meet setup time in a single clock cycle. However, we can also specify that some data is captured only after a specified number of clock cycles. Till then, the data at the capturing flop will not be used. Of course your circuit should be designed in such a way for this kind of behaviour to be valid. This is usually a large combinational block between two registers. It is important to specify the multicycle paths to synthesis and place&route tools, as the tools will otherwise try to fix these paths.

This timing exception is specified by the SDC command “set_multicycle_path”. This lets you specify the number of clock cycles required for the path.

Let us take the timing path from the previous post setup and hold. Let us say the datapath requires 3 clock cycles.

The clock diagram is given below. Assume the launch is at edge 1 of CLK.

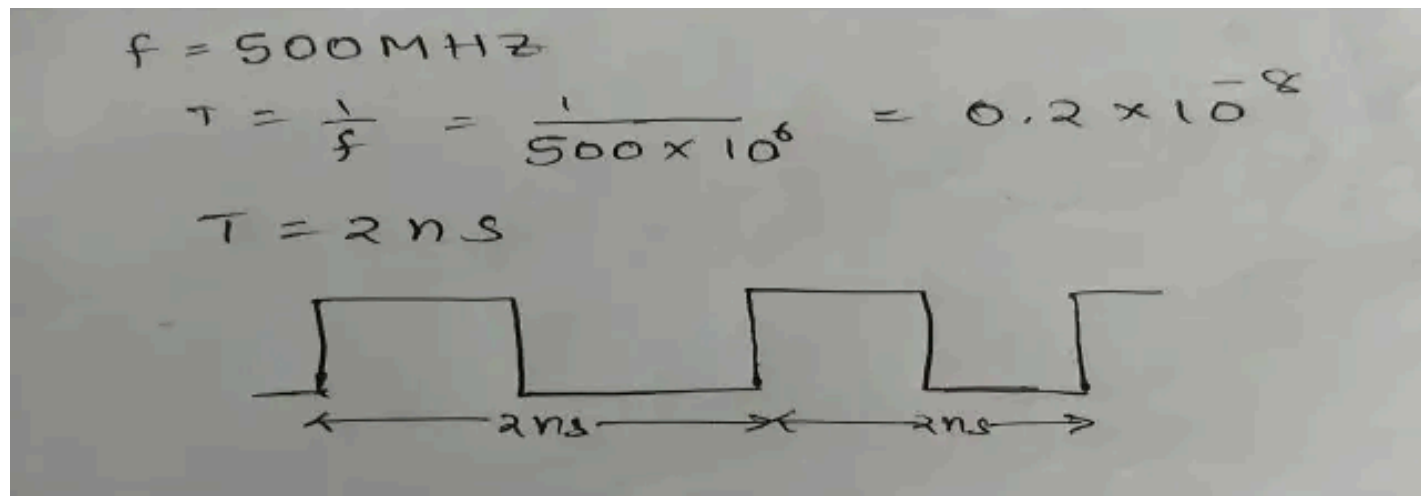


Once you have this specification, the STA tool takes the clock edge 4 as the capturing edge for FF2. By default, the hold is always checked one clock edge prior to setup edge. Hence the hold will be checked at edge 3. If you want the hold check to be done at another edge, say the launch edge itself, a set_multicycle_path -hold should also be given along with the setup specification. set_multicycle_path 1 -hold -from FF1/CP -to FF2/D will move the hold edge by one clock cycle from the default hold edge. i.e. to 2. set_multicycle_path 2 -hold will move the hold checking edge 2 cycles from the default hold edge. i.e. to clock edge 1, which is the default hold edge without any set_multicycle_path specified .

60. Draw a clock waveform. What is the time period of the clock with 500MHZ frequency?

Ans: If frequency is 500MHZ,

$$T=2\text{ns}$$



By Chandu:

1. Write a perl program to print sum of the digits from 1 to 10

A. For(\$i=0;\$i<=10;\$i=\$i+1)

```
{
    $sum=$sum+1;
}
```

Print “ the total sum is : \$sum”;

2. What are all the fixing methods for setup and hold violations

A. Setup:

- Upsizing the cells
- Replace buffer with two inverters
- HVT to LVT
- If the net delay is more than break the net and insert the buffer
- Pin swapping
- Pulling the launch and pushing the capture
- Cloning

Hold:

- Inserting the buffers
- Downsizing the cells
- LVT to HVT
- Pushing the launch and pulling the capture

3. What is OCV

A. OCV – On Chip Variation

The variations are mainly caused by the three factor. They are

- Process variation
- Voltage variation
- Temperature variation

Process variation: The process of fabrication includes diffusion, drawing out of metal wires, gate drawing etc. The diffusion density is not uniform throughout wafer. Also, the width of metal wire is not constant. Let us say, the width is 1um +- 20 nm. So, the metal delays are bound to be within a range rather than a single

value. Similarly, diffusion regions for all transistors will not have exactly same diffusion concentrations. So, all transistors are expected to have somewhat different characteristics.

Voltage variation: Power is distributed to all transistors on the chip with the help of a power grid. The power grid has its own resistance and capacitance. So, there is voltage drop along the power grid. Those transistors situated close to power source (or those having lesser resistive paths from power source) receive larger voltage as compared to other transistors. That is why, there is variation seen across transistors for delay.

Temperature variation: Similarly, all the transistors on the same chip cannot have same temperature. So, there are variations in characteristics due to variation in temperatures across the chip.

4. What is AOCV and POCV

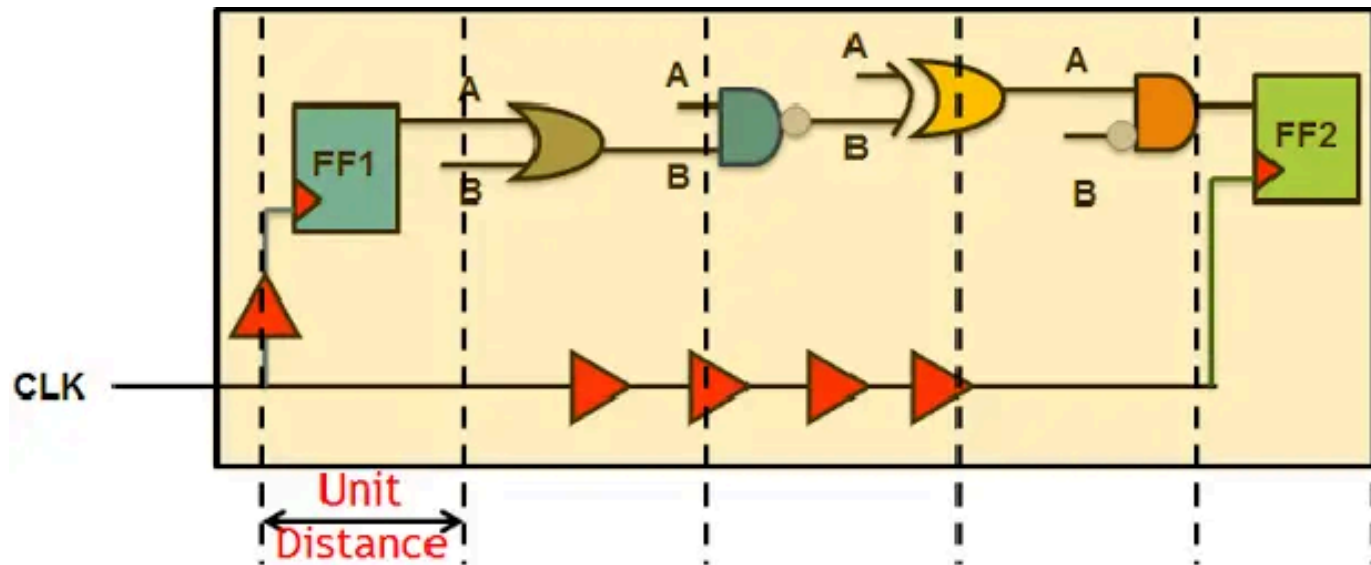
A. AOCV – Advanced On Chip Variation

This methodology hinges on three major concepts

Cell type: variations should take into account the cell type. Surely an AND gate and an OR gate can't exhibit the same variation pattern. Nor could an AND3X and an AND6X cell. The impact of variation should be calculated for each individual cell.

Distance: As the distance in x-y coordinates increase, the systematic variation would increase and we might need to use a higher derate value to reflect the uncertainty in timing analysis to mitigate any surprises on silicon.

Path Depth: If within a given distance, path depth is more, the impact of systematic variations would be constant, but the random variations would tend to cancel each other, Therefore as the path depth increases (within the same uniform distance), the AOCV derates tend to decrease.



While performing the reg2reg timing analysis, AOCV methodology finds the bounding box containing the sequential, clock buffers between two sequentials and all the data cells. Now within a unit distance, if the path depth increases, the AOCV derate decreases due to cancelling of random variations. However, if the distance increases, AOCV derates increases due to increase in the systematic variations. These variations are modeled in form of LUT.

Path Depth →

Setup	2	4	6	8
10	1.10	1.08	1.06	1.04
20	1.12	1.10	1.08	1.05
30	1.14	1.12	1.11	1.09
40	1.16	1.15	1.14	1.13

Distance ↓

POCV – Parametric On Chip Variation

POCV uses a statistical approach, but it doesn't do a full SSTA analysis. Instead, it calculates delay variation by modeling the intrinsic cell delay and load parasitic (line resistance, line capacitance, and load capacitance) to determine both the mean and "sigma" (variation) of a logic stage. The cell delay can be further broken into an n-channel component and a p-channel component. They then assume that all the cells along a path have the same mean and sigma.

This means that a given path doesn't have to be analyzed stage-by-stage; the number of stages can be counted, with the basic stage delay mean and sigma then used to calculate the path delay and accumulated variation. They claim that

this keeps the run times down to just over what standard STA tools require, far faster than SSTA. They also claim speedier execution and greater accuracy than AOCV, and no derating tables are required.

POCV is a technique that has been proposed as a means of reducing pessimism further by taking elements of SSTA and implementing them in a way that is less compute-intensive.

POCV provides the following:

- Statistical single-parameter derating for random variations
- Single input format and characterization source for both AOCV and POCV table data
- Non statistical timing reports
- Limited statistical reporting (mean, sigma) for timing paths

5. How did you open the prime time

A. pt_shell

6. what will you ask pd team in order to do STA

A. SPEF
DEF

7. If PD team has generated 10 spefs after routing, which spef will you ask for?

A. Worst RC corner spef

8. PD inputs

A.

- lib (timing, functionality, power)
- Lef (physical info)
- V (logical connectivity)
- Sdc (clock definitions)
- Cpf/upf (consists of power domain info)
- Cap table (RC values for every net)

9. What is Netlist and what type of information contains it

A.

- logical connectivity information
- Design name,

- Hierarchy information,
- Modules information,
- Io ports,
- Instances,
- Input and output pins

10. What is Lib and what type of information contains it

A.

- Timing, power and cell functionality information
- Power supply information
- Operating conditions
- PVTs
- Look up tables
- Cell area, functionality, timing, power

11. What is Lef and what type of information contains it

A. Physical information of the metals, cells and macros.

Tech. lef:

- layer name,
- Type,
- Pitch,
- Width,
- Spacing,
- Area,
- Resistance,
- Capacitance,
- Thickness,
- Edge capacitance,
- Min density
- Max density
- Antenna area ratio
- Current density
- Vias,
- Via rules

Cell/Macro lef:

- Name,

- Class,
- Origin,
- Size,
- Symmetry
- Pins direction, use, antenna gate area

12. What is Sdc and what type of information contains it

A.

Clock definitions:

- Create clock,
- Create generate clock,
- Virtual clock

I/O delays:

- Input delay
- Output delay

DRVs:

- max tran,
- max cap,
- max fanout

Timing exceptions:

- false path,
- multicycle path,
- max delay,
- min delay
- case analysis
- uncertainty

13. What are all the sanity checks

A.

Library checks:

- missing cells
- Mismatched pins
- Duplicate cell name

Design checks:

- combinational loops

- Assign statements
- Multidrive nets
- Floating inputs
- Tristate buffers

Constraint checks:

- All flops are clocked are not
- no unconstrained paths
- IO delays

14. What are all the commands used in P&R flow

A.

Inputs:

- set_init_verilog
- set_init_leffile
- set_init_mmmc
- set_init_pwrnet
- set_init_gndnet
- init_design

Sanity Checks:

- checkNetlist
- checkDesign -physicalLibrary
- checkDesign -timingLibrary

Floor Planning:

- checkPinAssignment
- addEndCap -preCap FILL8 -postCap FILL8
- addWellTap -cell FILL8 -cellInterval 40
- checkDesign -floorplan

Power Planning:

- globalNetConnect
- addRing
- addStripe
- sRoute

Placement:

- placeDesign
- setPlaceMode
- optDesign –preCTS
- report_timing

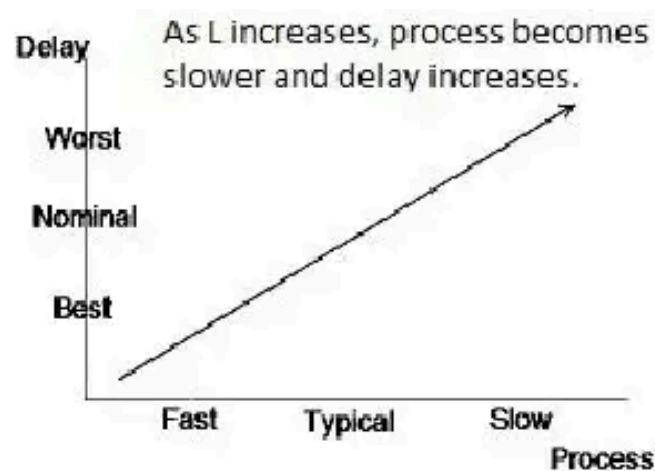
CTS:

- createClockTreeSpec
- clockDesign
- optDesign –postCTS –hold

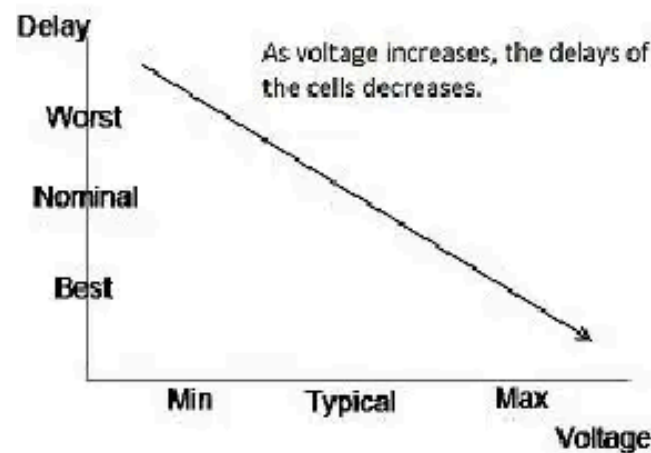
15. What do you mean by PVT conditions and how cell delays varies with PVT

A. PVT – Process Voltage and Temperature.

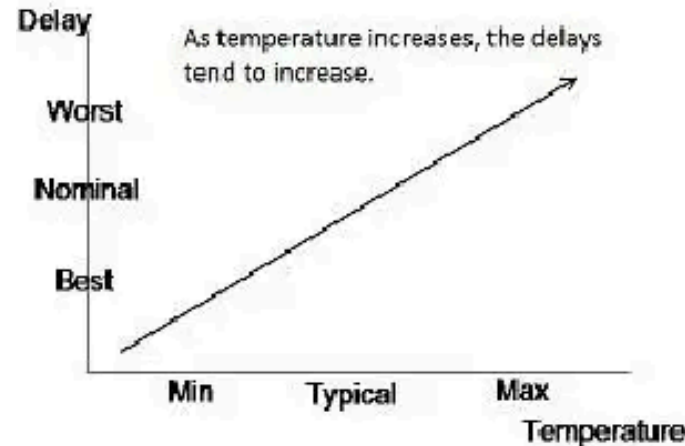
Process: You must have heard people talking in terms of process values like 90nm, 65nm, 45nm and other technology nodes. These values are characteristic of any technology and represent the length between the Source and Drain of a MOS transistor that you might have studied in your under-grad courses. While manufacturing any die, it has been seen that the dies that are present at the center are pretty accurate in their process values. But the ones lying on the periphery tend to deviate from this process value. The deviation is not big, but can have significant impact on timing.



Voltage: The voltage that any semiconductor chip works upon is given from outside. Recall while working on breadboards in your labs, you used to connect a 5V supply to the Vcc pin of your IC. Modern chips work on very less voltage than that. Typically around 1V-1.2V.



Temperature: The ambient temperature also impacts the timing. Let's say you are working on a gadget in Siachen glacier where temperature can drop down to -40 degrees centigrade in winters and you expect your device to be working fine. Or maybe you are in Sahara desert, where ambient temperature is +50 degrees and your car engine temperature is +150 degrees and again you expect your chip to working fine. While designing, therefore, STA engineers need to make sure that their chip will function correctly in the temperatures between -40 to +150 degrees.



16. How do you know you have max cap violation

A. report_timing -all_violators

17. What is static power

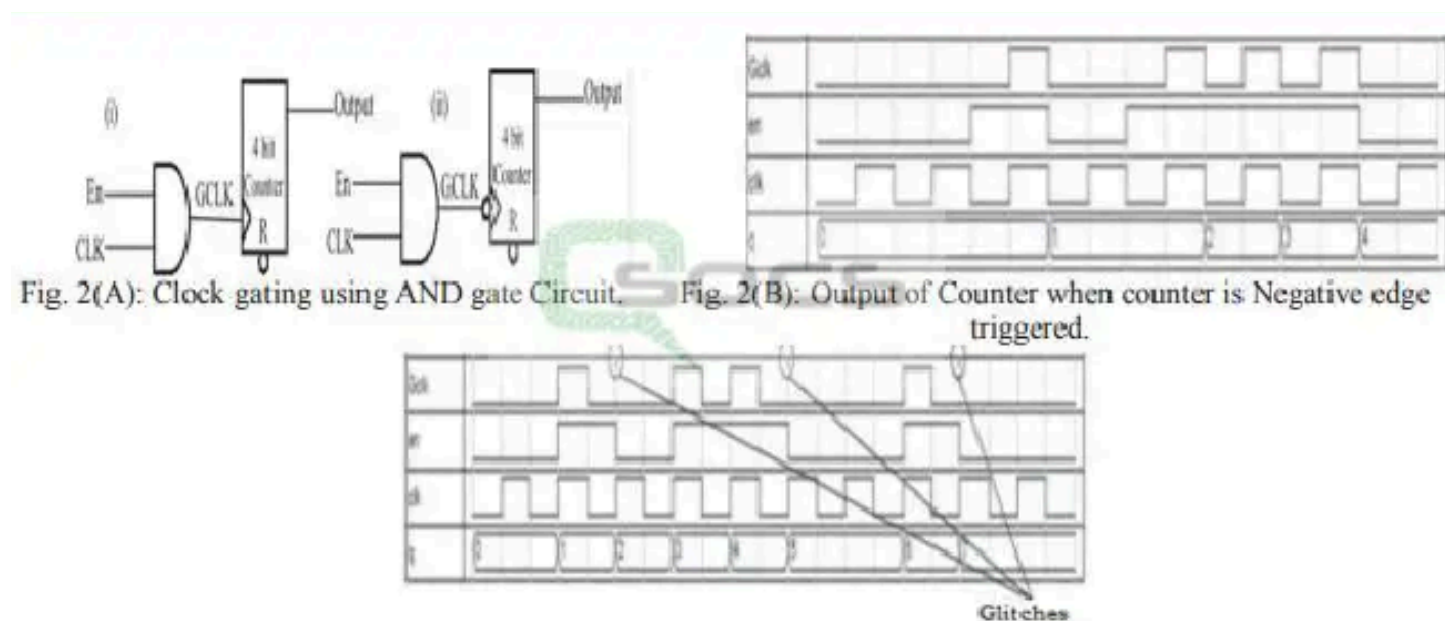
A. Analyzing the power and IR drop consumed in design when there is a constant current flow because of instances consuming average current in the design is call static power

18. What is dynamic power

A. Analyzing the power consumed and IR drop in design when the instances draws a transient current flow due to the instances switching

19. What is clock gating

A. Clock gating is a popular technique used in many synchronous_circuits for reducing dynamic power_dissipation. Clock gating saves power by adding more logic to a circuit to prune the clock_tree. Pruning the clock disables portions of the circuitry so that the flip-flops in them do not have to switch states. Switching states consumes power. When not being switched, the switching power consumption goes to zero, and only leakage_currents are incurred.



20. What is setup

A. Minimum amount of time the data must be stable before the active edge of the clock pluse.

21. What is cross talk

A. Crosstalk is any phenomenon by which a signal transmitted on one circuit or channel of a transmission system creates an undesired effect in another circuit or channel. Crosstalk is usually caused by undesired capacitive, inductive, or coupling from one circuit or channel to another.

22. Write a setup and hold equations

A. Setup:

$$\text{Arrival Time (AT)} = \text{launch clklatency} + \text{clk-q delay} + \text{combo delay}$$

Required Time (RT) = clk+ capture clklatency -lib setup time -(Uncertainty) + CRPR

Hold:

Arrival Time (AT) = launch clklatency + clk-q delay + combo delay

Required Time (RT) = clk+ capture clklatency + lib hold time+ Uncertainty – (CRPR)

23. What do you mean by aggressor net

A. Aggressor net creates impact to the neighboring nets

24. Difference between latches and flip flops

A.

Flip-flop	Latch
A flip-flop samples the inputs only at a clock event (rising edge, etc.)	A Latch samples the inputs continuously <i>whenever it is enabled</i> , that is, only when the enable signal is on. (or otherwise, it would be a wire, not a latch).
Flip-Flop are edge sensitive.	Latches are level sensitive.
Flipflop is sensitive to signal change and not on level. They can transfer data only at the single instant and data cannot be changed until next signal change.	Latch is sensitive to duration of pulse and can send or receive the data when the switch is on.
A flip-flop continuously checks its inputs and correspondingly changes its output only at times determined by clocking signal.	Latch is a device which continuously checks all its input and correspondingly changes its output, independent of the time determined by clocking signal.
It work's on the basis of clock pulses.	It is based on enable function input
It is a edge triggered, it mean that the output and the next state input changes when there is a change in clock pulse whether it may a +ve or -ve clock pulse.	It is a level triggered, it mean that the output of present state and input of the next state depends on the level that is binary input 1 or 0.

25. What are the disadvantages of latches over flipflops

A.

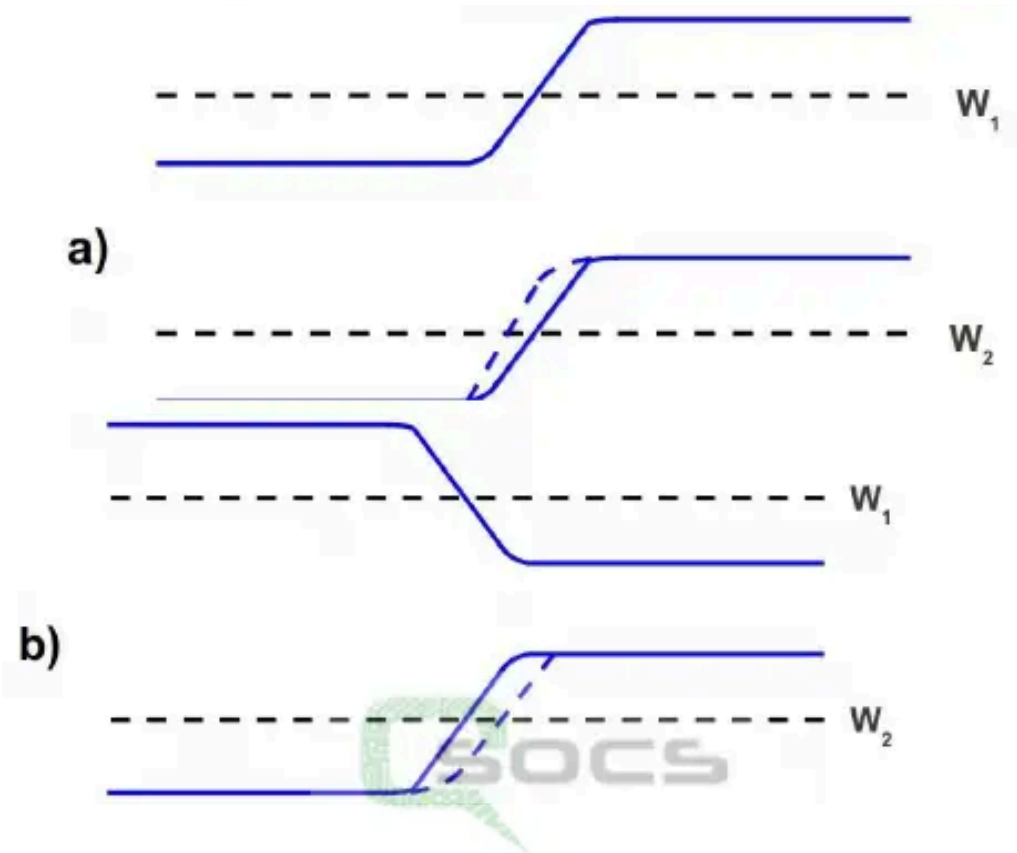
- Latch less predictable because there is more chance to affect to race conditions.
- Level sensitive devices and hence more chance of metastability.
- Analyzing of Latch circuits is difficult because of its level sensitive property.

26. What is cross talk noise and cross delay

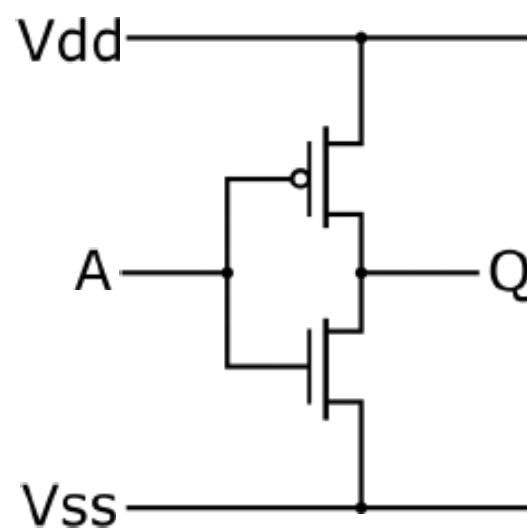
A. Cross talk noise: When w_2 is having a constant signal, and when w_1 signal is having a transition, it produces a spike at the w_2 signal. This glitch is called crosstalk noise. This is also referred to as bump violation. This glitch if wide and large enough, can get propagated through the logic, and can create a logical failure. The bump violations in clock network should be totally under control. Refer to the below figure for cross talk noise.



Cross talk delay: When w_2 signal is having a transition, this transition time can be affected by the signal transition at w_1 . If the transition at w_1 is in the same direction as w_2 signal transition, it will make w_2 signal transition slower. i.e., the net delays of the victim net will change with respect to the signal transition in the aggressor net, this should be taken care in the timing analysis for proper timing closure. But keep in mind, that the victim net will get affected, only if the transitions in both the nets happen in the same timing window. Also the delay change in the victim depends on the type of transition (rise or fall) in the aggressor net. See the below figures for more understanding.

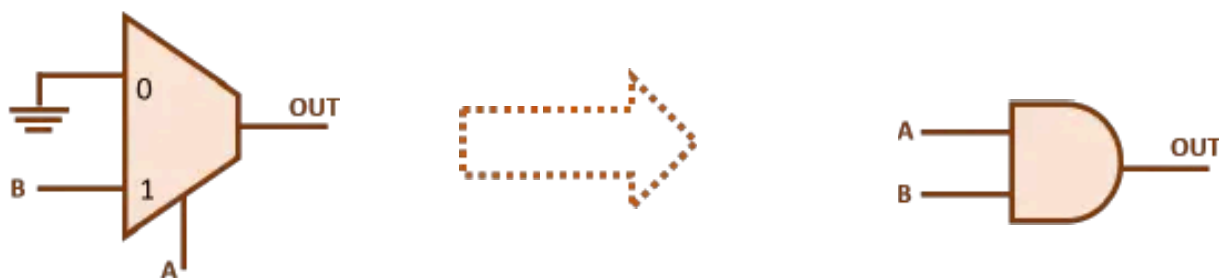


27. Draw the cmos inverter circuit

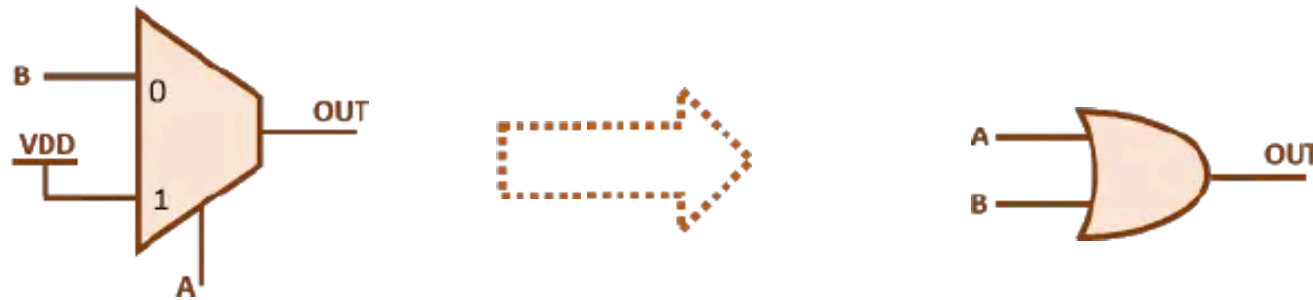


28. Draw 2X1 mux using AND OR gates and truth table

A. AND gate:



OR gate:



29. What do you mean by Mux and Demux

A. MUX: A multiplexer of 2^n inputs has n select lines, which are used to select which input line to send to the output. Multiplexers are mainly used to increase the amount of data that can be sent over the network within a certain amount of time and bandwidth. A multiplexer is also called a data selector. Multiplexers can also be used to implement Boolean functions of multiple variables.

DEMUX: A demultiplexer of 2^n outputs has n select lines, which are used to select which output line to send the input. A demultiplexer is also called a data distributor.

30. What is counter

A. Counter is a sequential circuit. A digital circuit which is used for counting pulses is known counter. Counter is the widest application of flip-flops. It is a group of flip-flops with a clock signal applied. Counters are of two types.

- Asynchronous or ripple counters.
- Synchronous counters.

31. Write a perl code to copy a one file into another file

```
A. Open(M1,"file1");
   Open (M2,">file2");
   @l=<M1>;
   Print M2 "@l \n";
```

32. What is signal integrity

A. Signal integrity or SI is a set of measures of the quality of an electrical signal. In digital electronics, a stream of binary values is represented by a voltage (or current) waveform

33. Write the sample verilog code

```
1 //-----
2 // Design Name : encoder_using_if
```

```

3 // File Name : encoder_using_if.v
4 // Function : Encoder using If
5 // Coder : Deepak Kumar Tala
6 //-----
7 module encoder_using_if(
8 binary_out , // 4 bit binary output
9 encoder_in , // 16-bit input
10 enable // Enable for the encoder
11 );
12 //-----Output Ports-----
13 output [3:0] binary_out ;
14 //-----Input Ports-----
15 input enable ;
16 input [15:0] encoder_in ;
17 //-----Internal Variables-----
18 reg [3:0] binary_out ;
19 //-----Code Start-----
20 always @(enable or encoder_in)
21 begin
22 binary_out = 0;
23 if (enable) begin
24 if (encoder_in == 16'h0002) begin
25 binary_out = 1;
26 end if (encoder_in == 16'h0004) begin
27 binary_out = 2;
28 end if (encoder_in == 16'h0008) begin
29 binary_out = 3;
30 end if (encoder_in == 16'h0010) begin
31 binary_out = 4;
32 end if (encoder_in == 16'h0020) begin
33 binary_out = 5;
34 end if (encoder_in == 16'h0040) begin
35 binary_out = 6;
36 end if (encoder_in == 16'h0080) begin
37 binary_out = 7;
38 end if (encoder_in == 16'h0100) begin
39 binary_out = 8;
40 end if (encoder_in == 16'h0200) begin
41 binary_out = 9;
42 end if (encoder_in == 16'h0400) begin
43 binary_out = 10;
44 end if (encoder_in == 16'h0800) begin
45 binary_out = 11;

```

```

46   end if (encoder_in == 16'h1000) begin
47       binary_out = 12;
48   end if (encoder_in == 16'h2000) begin
49       binary_out = 13;
50   end if (encoder_in == 16'h4000) begin
51       binary_out = 14;
52   end if (encoder_in == 16'h8000) begin
53       binary_out = 15;
54   end
55 end
56 end
57
58 endmodule

```

34. How can you fix max cap violations

A.

- Upsizing
- Adding the buffer

35. What is synthesis and what are the inputs to synthesis

A. Synthesis is the process of transforming your HDL design into a gate-level netlist, given all the specified constraints and optimization settings.

Inputs:

- Lib
- RTL
- Sdc

36. What is STA

A. STA is a method of validating the timing performance of a design by checking all possible paths for timing violations under worst case conditions.

37. Create a clock with 10ps period and rise edge 4ps and fall edge 8ps

A. `create_clock [get_ports{clk}] -name pclk -period 10 -waveform{4 8}`

38. what are mandatory things for generate clock

A. .

- Source master pin
- Divided by /multiple by factor

39. Write a command for generating clock

A. `Create_generated_clock -name clk -source PLL -divided_by 2 [get_pins uff0/q]`

40. What type of library did you use in your design

A. .

- Fast Lib
- Slow Lib

41. What are asynchronous circuits

A. The circuit in which the change in the input signals can affect memory elements at any instants of the time is call asynchronous circuit

In this circuit, clock is absent and hence the state changes can occur according to delay time of the logic.

42. Why do you need the virtual clock

A. To address the IO delays

43. What is recovery and removal

A. Recovery:

The recovery time is the minimum time that an asynchronous input is stable after being de-asserted before next active clock edge

Removal:

The removal time is the minimum time after an active clock edge that the asynchronous pin must remain active before it can be de-asserted

44. What does report_constraints –all_violators contain

A. It show all the violations

- Max_transition
- Max_fanout
- Max_capacitance
- Setup
- Hold

45. In sanity checks if we have floating inputs, can we move forward

A. No, we don't move forward because the floating inputs can take any input value from the neighboring switching nets. Then the undesired output will appear. so, we don't move forward if there is any floating input.

46. While giving the inputs to the floorplan, which LEF(tech lef or cell lef) we have to give first

A. First, we give the tech lef. Whenever the tech lef is loaded after all other inputs are loaded

47. Is .lib is mandatory input for floorplan, and what information you take from lib

A. No, .lib is not mandatory for floorplan

48. What are the floorplan inputs

A. Lef and .v

49. Guidelines of floorplan

A.

- IO ports must be placed.
- communication between macros and io ports
- macro to macro communication should be checked
- macros should be placed around the core boundary only
- macro pins should be towards the core area only
- crisscrossing should not be there between macros
- proper channel spacing should be given between the macro
- spacing between macros should be covered with the macros

50. what is overshoot glitch and undershoot glitch

A. There is still a glitch which takes the victim net voltage above its steady high value. Such a glitch is called an overshoot glitch. Similarly, a falling aggressor when coupled to a steady low victim net causes an undershoot glitch on the victim net.

51. What are the mandatory things for creating clocks

A.

- Period
- Source

52. Why do you required setup checks and hold checks

A.

- To meet the timing
- Proper functionality of the device

53. If I increases power supply, what happen to cell delay

A. The cell delay will decrease

54. If I increase my metal width, what happens to its height

A. The height remains same

55. Why we are using different metal width for different layers

A. For supply the different currents inside the design. Top metal layers have high current flow and low metal layers have low current flow

56. What happens if in my design have a same metal width for all 9 layers

A. If the width is high the current flow is high. Due to high current flow standard cells will damage.

If the width is low then we see more IR drop

57. How cell delay depends on channel length

A. If the channel length is more then the cell delay is more and vice versa

58. If top level guys had not mentioned the aspect ratio, then how will you take the core area

A. $(\text{Std.cell area} + 12\%(\text{PD overload}) + \text{Analog IP} + 12\%(\text{Analog to digital space}) + \text{Macros} + 4\%) / \text{Utilization}$

By Madhu:

213. Based on what we will place the macros?

A. Hierarchy, macro to macro communication, macro to IO communication and guidelines.

214. If macros are placed at the center, what are the problems we get?

A. Congestion, net lengths will be more due to detouring, timing issues, IR drop.

215. How will you give spacing between macros?

A. Spacing = (no. of pins * pitch of routing layer)/no. of available layers in the preferred direction

216. To avoid congestion in between macros, what are the cares we take?

A. proper channel spacing should be provided between macro to macro and the halo should be provided around the macro, there should not be crisscrossing between macro to macro.

217. What are the types of blockages?

A. placement blockages:

1. Hard blockage: It will not allow any cells inside the blockage.

2. Soft blockage: It will not allow std cell but it will allow buffers and inverters during optimization.

3. Partial blockage: It will allow only certain specified percentage of cells.

Routing blockage: It will allow only some specified metal layers inside the blockage.

218. If we place soft blockage in between the macros it will allow the buffers and inverters, in that, then how will the power will get to that cells?

A. Stripe should be added to get the power to those cells.

219. What are all the targets after power plan?

A. Checks after power planning:

1. Connectivity should be verified.

2. Any open, floating or dangling nets in the design.

220. Inputs of placement?

A. Design database upto power plan, lib, sdc, scandef

221. What are contents in SDC?

A. Sdc: clock definitions: create clock, create generate clock, virtual clock

Input delay

Output delay

DRVs: max tran, max cap, max fanout

Timing exceptions: false path, multicycle path, max delay, min delay

222. What is multicycle path and false path?

A. The combinational data path between two flip-flops can take more than one clock cycle to propagate through the logic. In such cases, the combinational path is declared as a multicycle path.

Some certain timing paths are not real (or not possible) in the actual functional operation of the design. Such paths can be turned off during STA by setting these as false paths. A false path is ignored by the STA for analysis.

223. At the placement stage what are all the different types of congestion you see? And how you overcome that congestion in your design?

A. 1. Using hard blockages in case of congestion due to notches.

2. Using partial blockage, in case of high cell density.
3. Using cell padding, in case of congestion due to pin density.

224. Explain different techniques to overcome the congestion?

A. same as above

225. At placement stage, if we have the utilization of 95%, can we move forward and why?

A. No, we should not proceed with that utilization. We have to check the reason for the jump in the utilization number.

226. What are the timing checks we do after placement?

A. DRVs and setup checks

227. If the design had violated by 500ps of setup then will you move forward and why?

A. No, we have to fix the setup violation. If we proceed at this stage by the next stage the setup will be violated more there may not be chance to fix.

228. What are inputs of CTS?

A. placement database, sdc, lib, spec.

229. What exactly the spec will contain?

A. Clock grouping,
Through pin,
Exclude pin,
Clocks,
Period,
Sink pin,
Min. delay,
Max. delay,
Min. skew,
Clock buffers and inverters,
Routing layers,
NDRs



230. What is the use of clock groups in the spec file?

A. clock groups that are mutually exclusive or asynchronous with each other in a design so that the paths between these clocks are not considered during the timing analysis.

231. What are the targets of CTS?

A. balanced skew and minimum insertion delay.

232. What is the skew?

A. Skew is the difference in timing between two or more signals, maybe data, clock or both. Clock skew is the difference in arrival times at the end points of the clock tree. Clock latency is the total time it takes from the clock source to an end point.

Skew is the difference between capture flop latency and launch flop latency.

233. Explain setup and hold with equation.

A. Setup: The minimum time before the active clock when the data input must remain stable is called the setup time.

Setup time = Required Time - Arrival Time

Required time = Clock period + capture clock latency - lib setup - setup uncertainty

Arrival time = launch clock latency + C to Q delay + combo delay

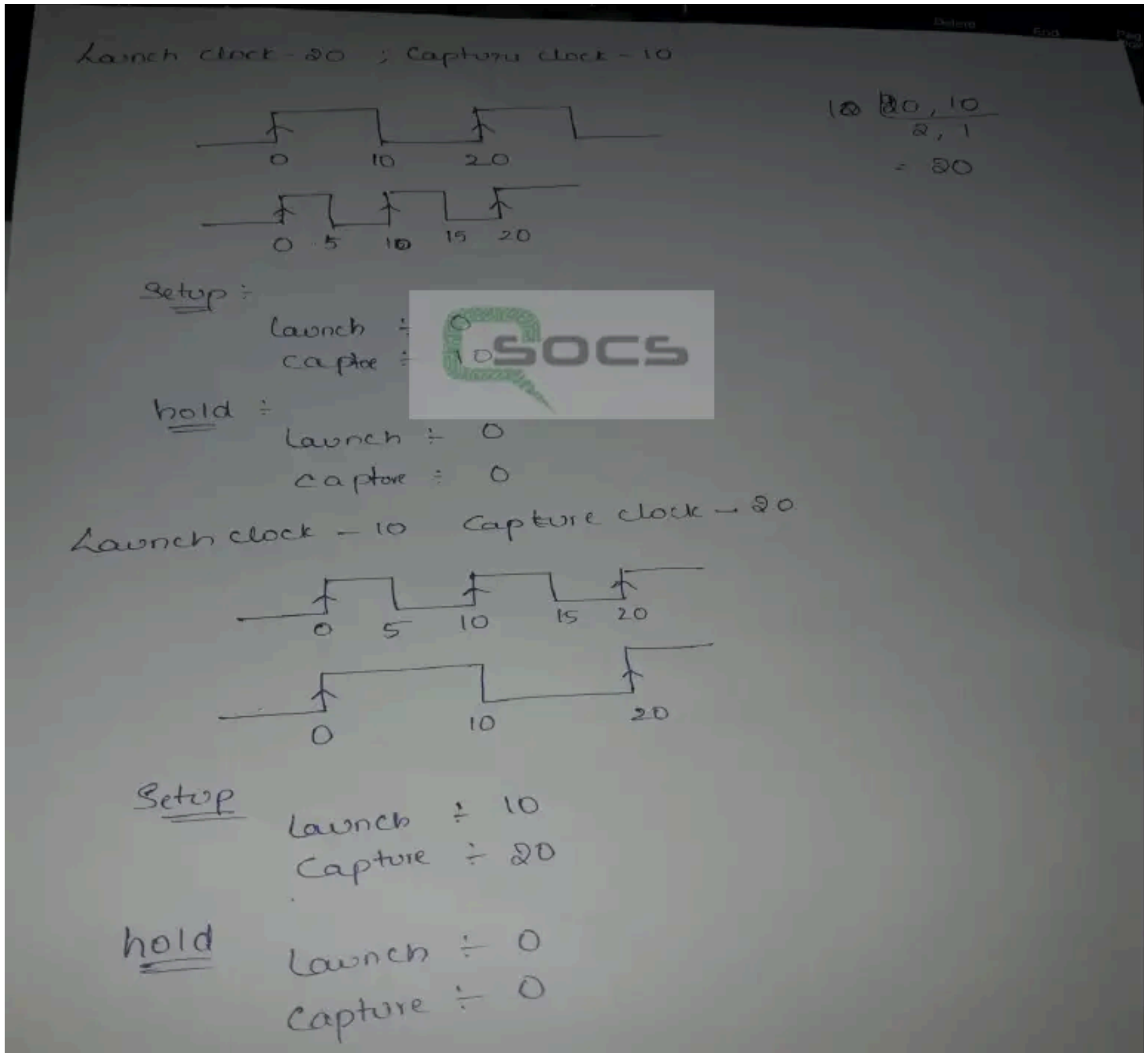
Hold: The minimum time the data input must remain stable just after the active edge of the clock.

Hold time = Arrival Time - Required Time

Required time = Capture clock latency - lib setup - setup uncertainty

Arrival time = launch clock latency + C to Q delay + combo delay

234. If launch clock is 20ns and capture clock is 10ns then where do we check the setup and hold? and vice versa?
A.



235. How to fix the setup and hold violations?

A. setup:

- upsizing,
- Vt swapping (HVT to LVT)
- Pushing capture flop (adding delay)
- Pulling launch flop (reducing delay)

Hold:

- downsizing

Vt swapping (LVT to HVT)
Pulling capture flop (reducing delay)
Pushing launch flop (adding delay)
Buffering for net delays

236. What is the resistor, capacitor and current formulae?

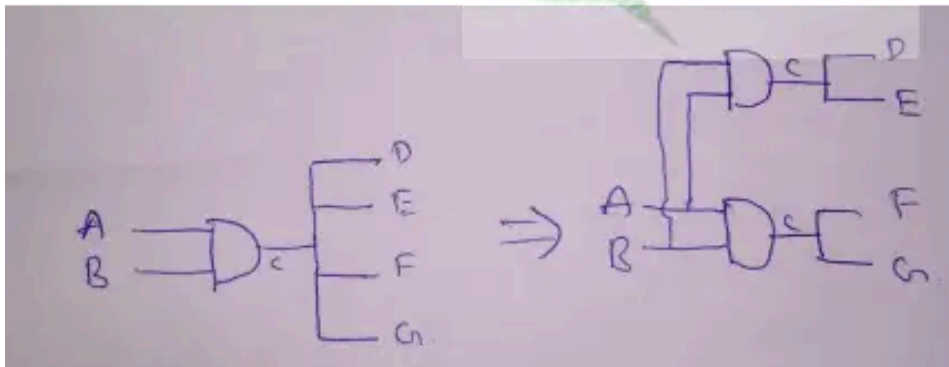
- A. Capacitance, $C = \epsilon A/d$,
Where, ϵ is the permittivity, A is area and d is distance.
Resistance, $R = \rho l/A$,
Where, ρ is resistivity, l is length and A is area.
Current, $I = V/R$,
Where, V is voltage and R is resistance.

237. What is antenna violation? On what basis antenna violation will affect the cell.

- A. It is also called “Plasma induced gate effect damage”. It will occur at manufacturing stage. It is a gate damage that can occur due to charge accumulation on metals and discharge to a gate through gate oxide. These are normally expressed as an allowable ratio of metal area to gate area greater than allowable area.

238. What is cloning?

A.



239. On what tool and technology, you have worked? Have you ever seen PT and calibre tools?

241. What happens if we have setup and hold violations in our design?

- A. setup and hold timings are to be met in order to ensure that data launched from one flop is captured properly at another and in accordance to the state machine designed. In other words, no timing violations means that the data launched by one flip-flop at one clock edge is getting captured by another flip-flop at the desired clock edge. If the setup check is violated, data will not be captured properly at the next clock edge. Similarly, if hold check is violated, data intended to get captured at the next edge will get captured at the same edge. Moreover, setup/hold violations can lead to data getting captured within the setup/hold window which can lead to metastability of the capturing flip-flop. So, it is very important to have setup and hold requirements met for all the registers in the design and there should not be any setup/hold violations.

242. What will you do if your design has setup violation and how will you meet setup in such cases where there are no margins available?

- A. We can meet the setup violations by adding buffers and downsizing the buffers present in the clock path.

243. What is time borrowing?

- A. The time borrowing technique, which is also called cycle stealing, occurs at a latch. In a latch, one edge of the clock makes the latch transparent, that is, it opens the latch so that output of the latch is the same as the data input; this clock edge is called the opening edge. The second edge of the clock closes the latch, that is, any change on the data input is no longer available at the output of the latch; this clock edge is called the closing edge.

244. What are the concepts involved in STA?

A. modeling of cells, propagation delay, slew of waveform, skew between signals, Timing arcs, Timing paths, Clock domains,
Operating conditions.

245. What is signal integrity?

A. The ability of the signal to withstand the interference from the nearby signal.

246. What is aggressor net? Why it's called aggressor net?

A. The net in which high switching activity is performed is aggressor. It is the affecting net so that it is called as aggressor net.

247. Both nets are having same drive strength, but voltage is different, capacitance is occurred or not?

A. yes, the capacitance occurs due to different voltage levels.

248. Both nets are having same voltage, but voltage is same, capacitance is occurred or not?

A. No, the capacitance does not occur because the voltage levels are same.

249. Both nets having same voltage capacitance is occurred or not?

A. No

250. What is cross coupling capacitance? Why it occurs?

A. The capacitance formed between two different nets in which one of the net is having high switching activity and other net is containing low level signal. The capacitance is formed because of different voltage levels in the nets and different switching.

251. What is dielectric? Do know dielectric formula?

A. A dielectric (or dielectric material) is an electrical insulator that can be polarized by an applied electric field. When a dielectric is placed in an electric field, electric charges do not flow through the material as they do in an electrical conductor but only slightly shift from their average equilibrium positions causing dielectric polarization. Because of dielectric polarization, positive charges are displaced in the direction of the field and negative charges shift in the opposite direction.

Dielectric constant, $\kappa = C/C_0$

Where, C is value of the capacitance of a capacitor filled with a given dielectric

C_0 is the capacitance of an identical capacitor in a vacuum

253. What is cross talk? Explain in detail.

A. The interference of the high switching signal on the low level signal due to the formation of coupling capacitor between the

nets. The net in which high switching activity is there and affecting the idle net is called aggressor net. The net which is having low switching activity and affected by aggressor is called victim net.

When aggressor net is switching and victim net is idle, we may get cross talk noise.

When aggressor net and victim net both are switching, we may get cross talk delay.

254. How to avoid crosstalk?

A. Applying NDRs, downsizing aggressor, upsizing victim, shielding victim net.

257. What is data path?

A. The path between Q pin of the launch flop and D pin of the capture flop is called data path.

258. What is OCV?

A. The process and environmental conditions may not be uniform across the different portions of the chip. Due to process variations, identical transistors may not have similar characteristics in different portions of the chip.

259. What is process?

A. Process variation is the naturally occurring variation in the attributes of transistors (length, widths, oxide thickness)

when integrated circuits are fabricated. The amount of process variation becomes particularly pronounced at smaller process nodes ($<65\text{ nm}$) as the variation becomes a larger percentage of the full length or width of the device and as feature sizes approach the fundamental dimensions such as the size of atoms and the wavelength of usable light for patterning lithography masks.

260. Explain the corners.

A. Parasitics can be extracted at many corners. These are mostly governed by the variations in the metal width and metal etch in

the manufacturing process. Some of these are:

- *Typical*: This refers to the nominal values for interconnect resistance and capacitance.
- *Max C*: This refers to the interconnect corner which results in maximum capacitance. The interconnect resistance is smaller than at *typical* corner. This corner results in largest delay for paths with short nets and can be used for max path analysis.
- *Min C*: This refers to the interconnect corner which results in minimum capacitance. The interconnect resistance is larger than at *typical* corner. This corner results in smallest delay for paths with short nets and can be used for min path analysis.
- *Max RC*: This refers to the interconnect corner which maximizes the interconnect RC product. This typically corresponds to larger etch which reduces the trace width. This results in largest resistance but corresponds to smaller than typical capacitance. Overall, this corner has the largest delay for paths with long interconnects and can be used for max path analysis.
- *Min RC*: This refers to the interconnect corner which minimizes the interconnect RC product. This typically corresponds to smaller etch which increases the trace width. This results in smallest resistance but corresponds to larger than typical capacitance. Overall, this corner has the smallest path delay for paths with long interconnects and can be used for min path analysis.

By Nandini:

262. Tell me about STA?

Ans: It is a method of validating the timing performance of the design by checking all possible paths for timing violations under worstcase conditions.

Read lib

Read netlist

(Link the design and check any unresolved references and black box)

Read SDC

(check_timing(we should not get any unconstrained end points; clock notfound; No drive Assertion))

Read spef

(report_annotated parasitics)

Report timing

(we need to generate timing reports for all variable paths in the design)

Analysis

(After generating the reports we should analyse all the slack, setup, hold values)

263. Tell me about PD?

Ans: A. Inputs for PD are

1. lib (timing, functionality, power)
2. lef (physical info)
3. v (logical connectivity)
4. sdc (clock definitions)
5. cpf/upf (consists of power domain info)
6. cap table (RC values for every net)

B. Sanity checks

a) Library sanity checks (physical and logical library)

1. Missing cells and pins
2. Mismatched pins
3. Duplicate cell name

b) Design sanity checks (checkNetlist)

1. Floating pins
2. Combinational loops
3. Assign statements
4. Tristate buffers
5. Empty modules
6. Multi Driven nets

c) Constraint sanity checks

1. All flops are clocked or not
2. No unconstrained path
3. Input and Output delays

- Floorplan (defining core and die area based on Aspect ratio and Utilization factor and placing macros physical cells)
- Power Plan (supplying power from Pads to the Design)
- Placement (finding the appropriate location of the std cells)
- Place opt (to meeting area congestion and timing [setup])
- CTS (Distributing clock from clock port to clock pin)
- opt CTS (balancing skew and latency also DRC's to meet the timing[setup, hold])
- Routing (Connecting components physically and finding optimized net length with introduced net delays)
- opt Routing (Assigning the track for every net and fixing the timings)
- Signoff (meeting the all requirements timing, functionality, power, area and DRC.)

264. Inputs of STA?

Ans: 1. Lib
2. netlist
3. sdc
4. Spef

265. Inputs of PD?

Ans: Technology Related inputs:

1. .lib
2. Lef
3. captables

Design Related inputs:

1. .v
2. SDC
3. CPF

266. What is Setup? How you fix setup?

Ans: The time taken by the data to be stable before the clock edge called Setup.

How to fix setup :

1. Upsizing the cell.
2. Vt swapping.
3. Pulling the launch.
4. Pushing the capture.

267. what is hold? How u fix hold?

Ans: The time taken by the data to be stable before the clock edge called hold.

How to fix hold:

1. Downsizing the cell.
2. Vt swapping.
3. Pulling the capture.
4. Pushing the launch.
5. Inserting the buffer in the data path.

271. What is BlackBox?

Ans: It consists of input and output but we cannot see the inside functionality.

272. What are the commands using entire pd flow?

Ans:

FloorPlan:

1. floorPlan -site CoreSite -r 1 0.698983 1.0 1.14 1.0 1.0
2. editPin -fixedPin 1 -fixOverlap 1 -unit MICRON -spreadDirection clockwise -side Top -layer 4 -spreadType start -spacing 0.4 -start 4.337 750.8455 -pin {pin names}
3. addHaloToBlock 1 1 1 1 -allBlock
4. addEndCap -preCap FILL2 -postCap FILL2 -prefix ENDCAP
5. addWellTap -cell FILL8 -cellInterval 40 -fixedGap -prefix WELLTAP

PowerPlan:

1. for global net connection
globalNetConnect VDD -type pgpin -pin VDD -inst *
globalNetConnect VSS -type pgpin -pin VSS -inst *
globalNetConnect VDD -type tiehi -pin VDD -inst *
globalNetConnect VSS -type tielo -pin VSS -inst *

Placement:

1. placeDesign
2. for hard blockage (createPlaceBlockage -box <area>)
3. softblockage (createDensityArea <area> <percentage>)
4. placeDesign -incremental
5. optDesign -preCTS

CTS:

1. clockDesign
2. optDesign -postCTS

Routing:

1. RouteDesign
2. optDesign -postRoute

273. what are the commands using CTS level?

Ans: 1. clockDesign

2. optdesign-postCTS

3. report_timing

4. report_constraints -all _violators

274. Tell me ten ways to fix Setup and hold?

Ans: To fix SETUP:

1. Upsizing the cell.



2. Vt Swapping.
3. Pulling the launch.
4. Pushing the capture.
5. Logical Restructuring.
6. Pin Swapping.
7. Cloning.
8. If net delay dominates the cell delay :

Break the net and adding buffers.

To fix HOLD:

1. Downsize the cell
2. Vt Swapping
3. Pulling the capture
4. Pushing the launch
5. Insert the buffer in the data path

276. What is OCV and AOCV?

Ans: : OCV:

Minor changes in delays due to the variations in PVT conditions. As cell delays are varying we will apply a global derating factor then every cell having min and max delay. All the cells are applying with same derating factor.

AOCV:

Here we are applying a derating factor based on logical depth and distance. OCV is more pessimistic than AOCV so we are going for AOCV.

277. Create Clock Command?

Ans: `create_clock -name top -period 10 -waveform (0 5) -get_ports[scan clk]`

278. What is Duty cycle?

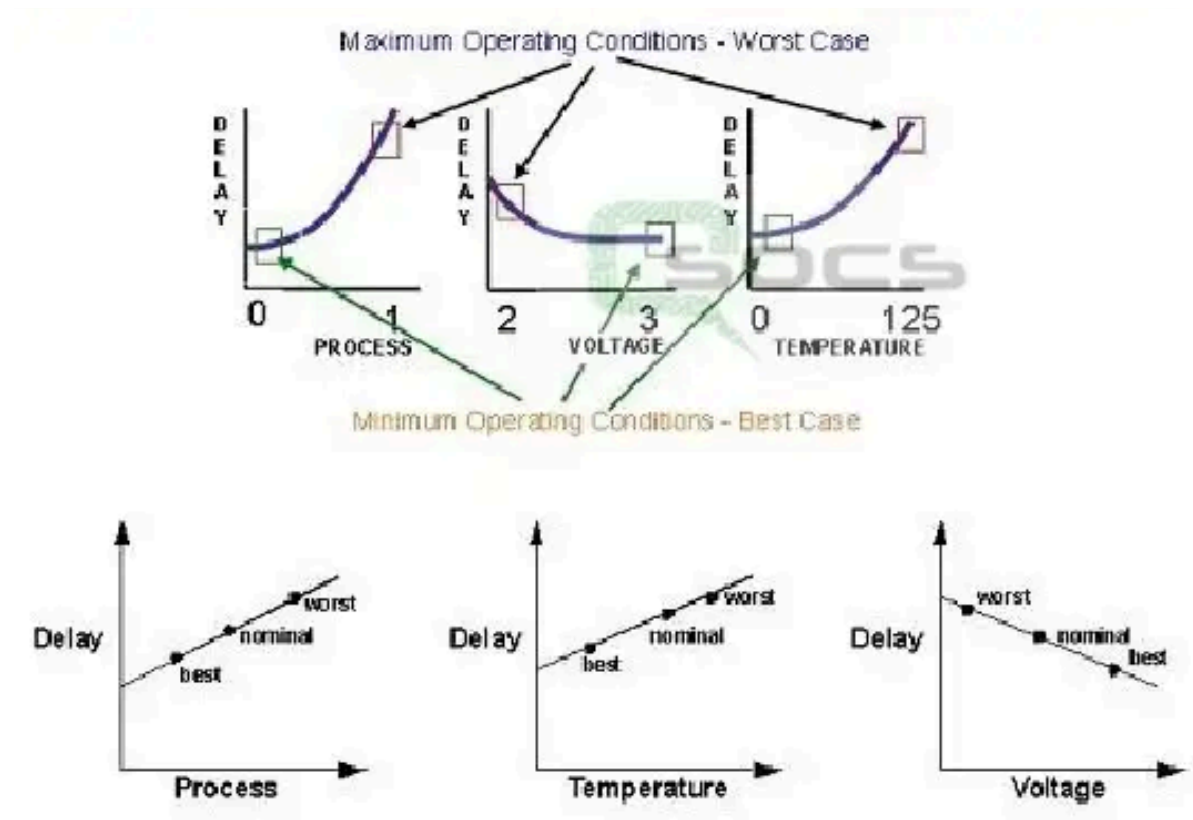
Ans: A **duty cycle** is the fraction of one period in which a signal or system is active known as Duty cycle.

$$\text{Duty cycle} = (\text{Ton} / (\text{Ton} + \text{Toff})) * 100$$

275. What is PVT?

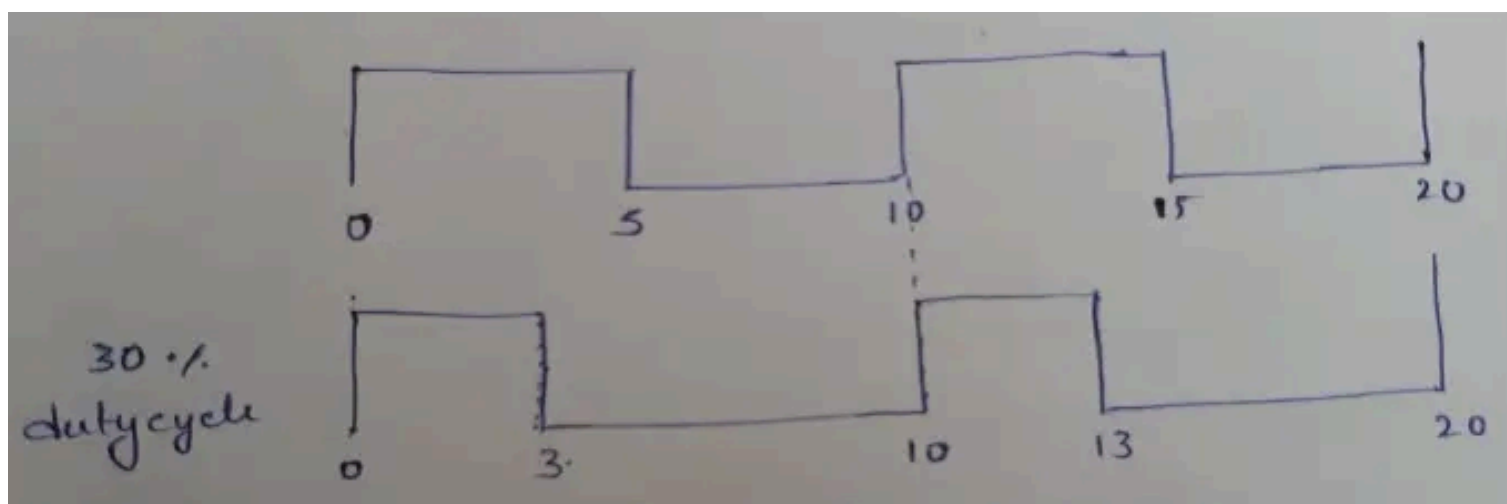
Ans: PVT- Process Voltage Temperature

- **Best case:** fast process, highest voltage and lowest temperature
- **Worst case:** slow process, lowest voltage and highest temperature
- **Normal case:** normal process, normal voltage, normal temperature



279. Make a waveform 30% duty cycle?

Ans:



280. How many spec files used in your project?

Ans: 6 spec files

281. Cell Delay Vs V_{th} ?

Ans: If temperature increases mobility decreases and V_{th} decreases.

If temperature decreases mobility increases and V_{th} increases.

In lower technologies temperature decreases delay increases because it depends on threshold voltage.

Because of Drain current i.e., I_d is inversely proportional to V_{th} if v_{th} increases I_d decreases then process is slow then delay also increases.

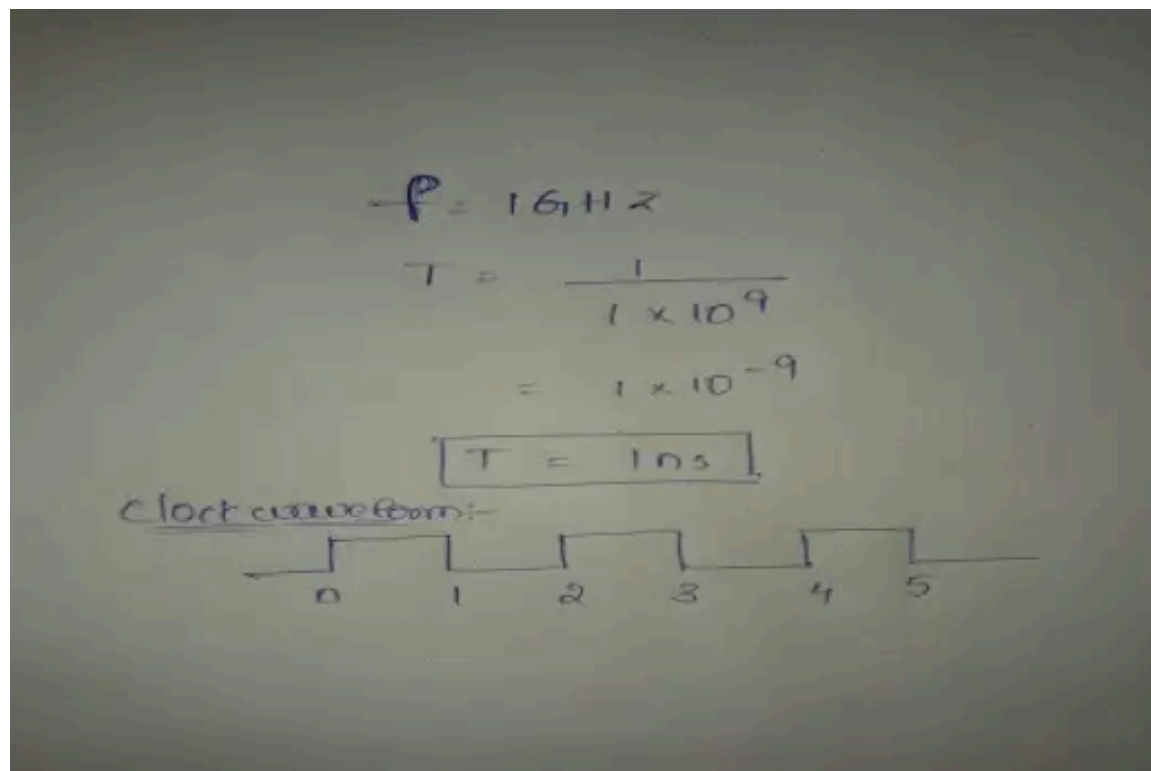
282. which is being used currently AOCV or POCV?

Ans: **POCV [Parametric On Chip Variation]:**

It uses a statistical approach, it calculates delay variation by modeling the intrinsic cell delay and parasitic of load which determines sigma and mean of a logic stage.

283. Draw a clock waveform. What is the time period of a clock with 1GHz frequency?

Ans:



284. What is netlist? What it contains?

Ans: Logical connectivity information between combinational and sequential cells known as Netlist. It contains

1. Module information
2. Hierarchical information
3. Input Output Notations
4. port information

5. Instance and net names

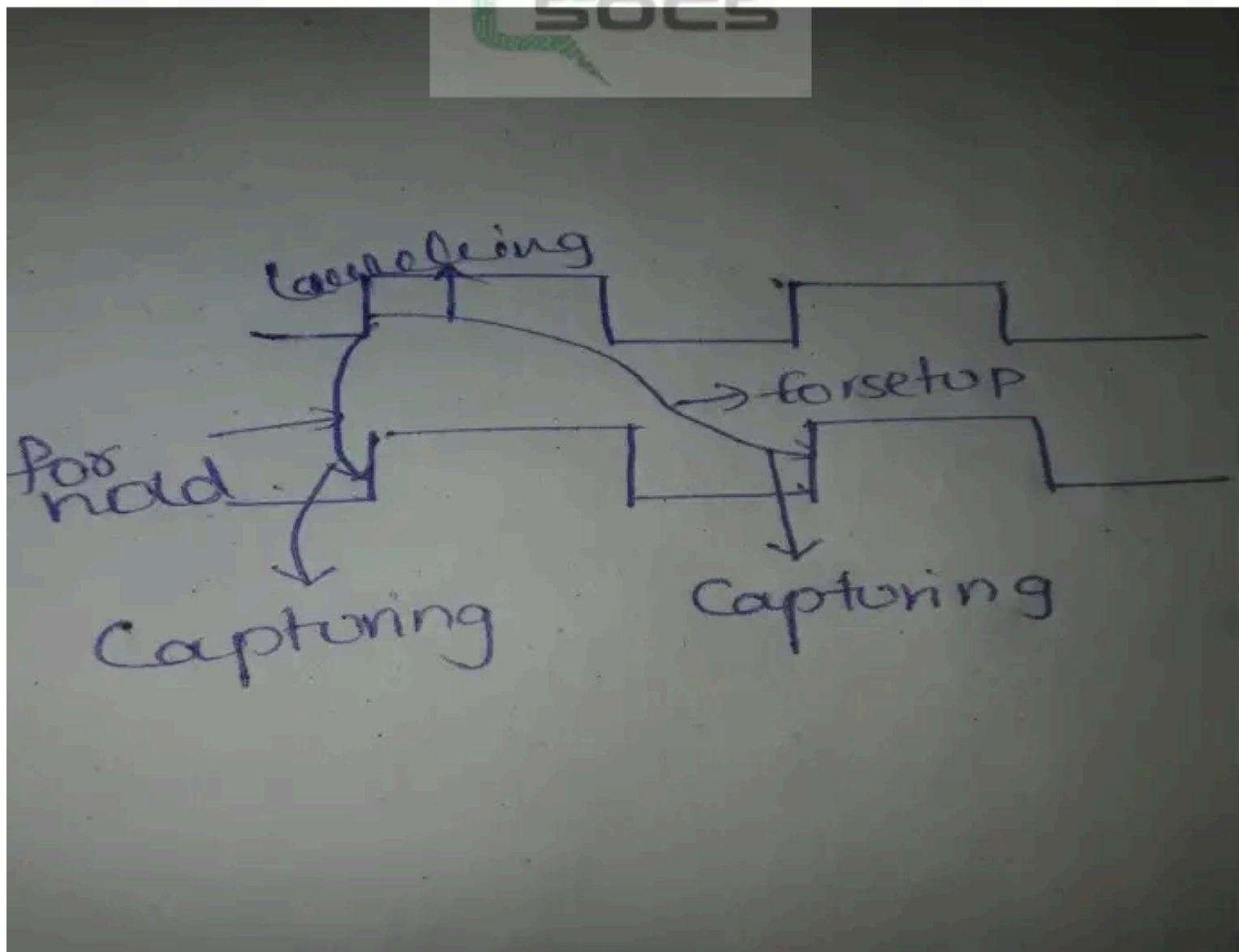
285. What .lib contains?

Ans: It contains

1. Power Information
2. Timing Information
3. cell Functionality
4. PVT

286. Where will the data get launched and captured?

Ans:



290. What is Setup Equation?

Ans: SETUP:

$$R.T - A.T$$

$R.T = \text{Clkperiod} + \text{Capture clock latency} - \text{lib.setup time} - \text{uncertainty}$

$A.T = \text{Launch Clock latency} + \text{Clk-Q delay} + \text{combinational path}$

HOLD:

$A.T - R.T$

$A.T = \text{Launch Clock latency} + \text{Clk-Q delay} + \text{combinational path}$

$R.T = \text{Capture clock latency} + \text{lib.hold time} + \text{hold uncertainty}$

291. If setup checks and hold checks not done what happened? If it necessary why?

Ans: Timing doesn't meet if we do not do setup and hold check.

292. Why NMOS is Called NMOS?

Ans: **N-type metal-oxide-semiconductor logic** uses N type field effect transistors MOSFETs to implement logic gates and other digital circuits. These NMOS transistors operate by creating an inversion layer in a p-type transistor body. This inversion layer, called the n-channel, can conduct electrons between n-type "source" and "drain" terminals. The n-channel is created by applying voltage to the third terminal, called the gate.

293. What is NMOS and PMOS ?

Ans: NMOS: N-type metal oxide semiconductor

A NMOS transistor is made up of n-type source and drain and a p-type substrate. When a voltage is applied to the gate, holes in the body (p-type substrate) are driven away from the gate. This allows forming an n-type channel between the source and the drain and a current is carried by electrons from source to the drain through an induced n-type channel.

PMOS: P-type Metal oxide Semiconductor

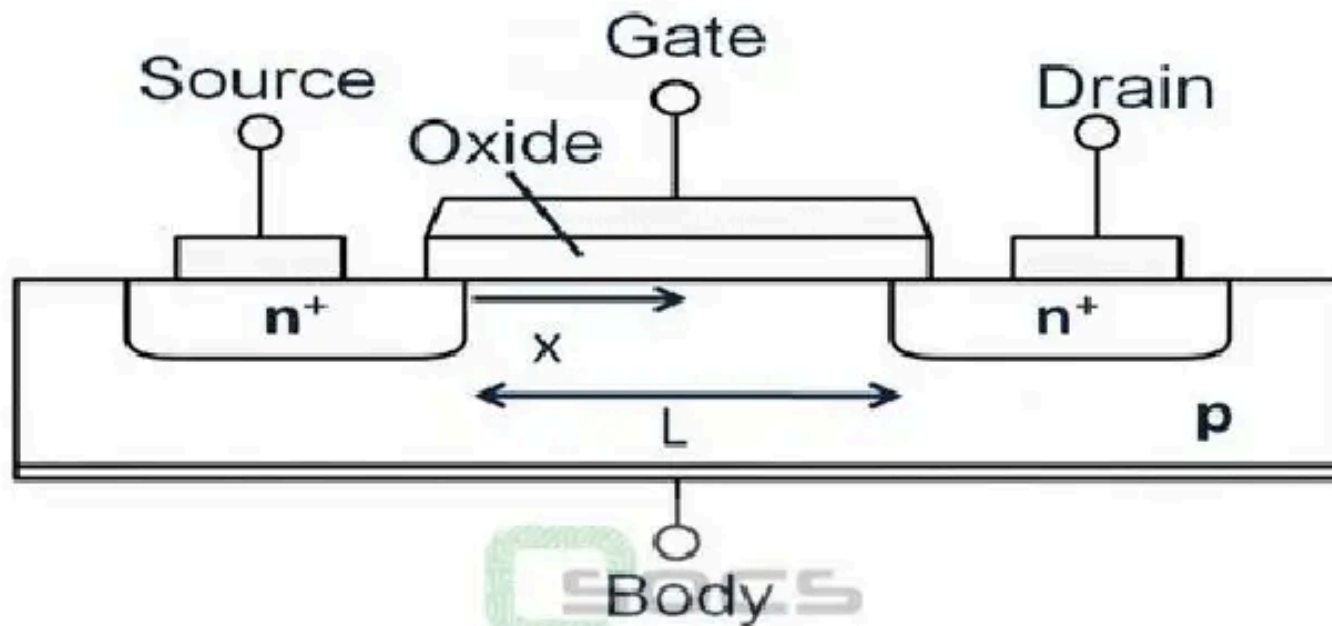
A PMOS transistor is made up of p-type source and drain and a n-type substrate. When a positive voltage is applied between the source and the gate (negative voltage between gate and source) a p-type channel is formed between the source and the drain with opposite polarities. A current is carried by holes from source to the drain through an induced p-type channel. A high voltage on the gate will cause a PMOS not to conduct, while a low voltage on the gate will cause it to conduct.

294. What is MOSFET? Draw the MOSFET Diagram?

Ans: MOSFET: Metal Oxide Semiconductor

The **metal-oxide-semiconductor field-effect transistor (MOSFET, MOS-FET, or MOS FET)** is a type of field-effect transistor (FET), most commonly fabricated by the controlled oxidation of silicon. It has an insulated gate, whose voltage determines the

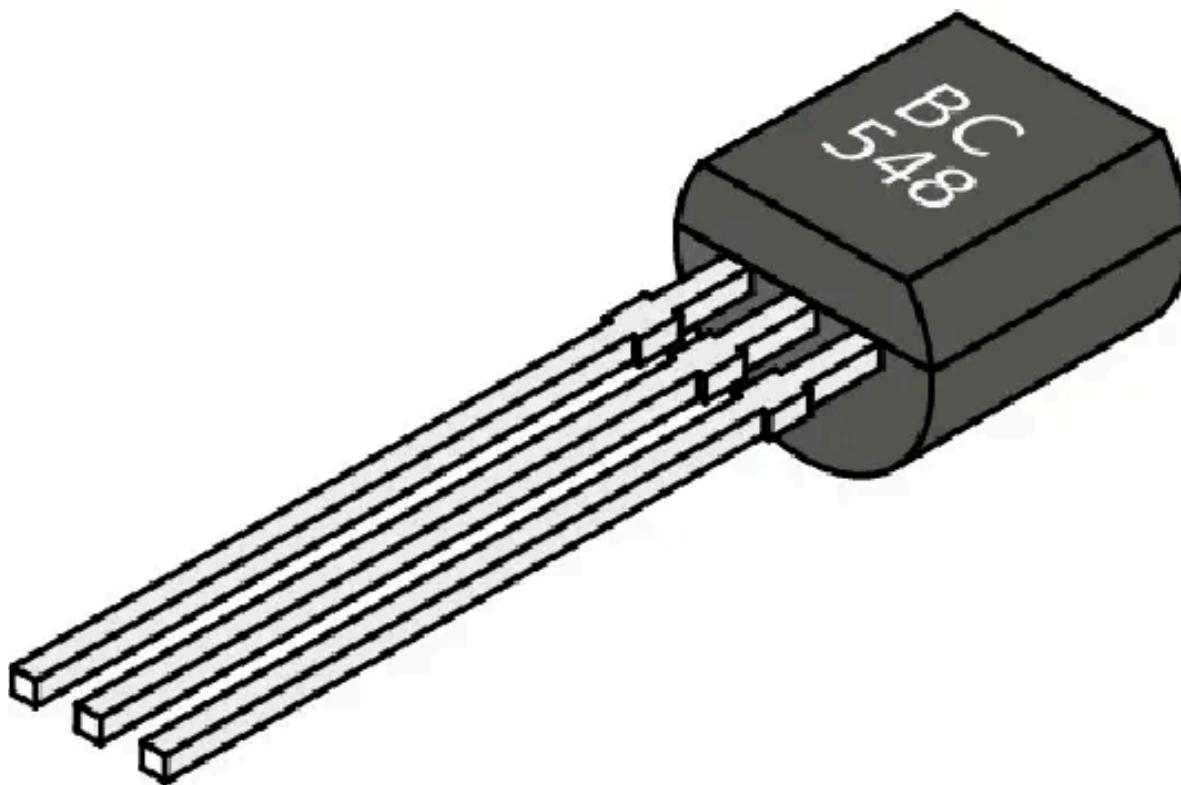
conductivity of the device. This ability to change conductivity with the amount of applied voltage can be used for amplifying or switching electronic signals.



295. What is Transistor?

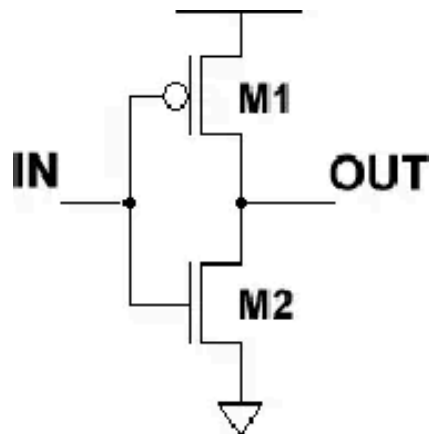
Ans: Transfer of Resistance

A **transistor** is a semiconductor device used to amplify or switch electronic signals and electrical power. It is composed of semiconductor material usually with at least three terminals for connection to an external circuit.



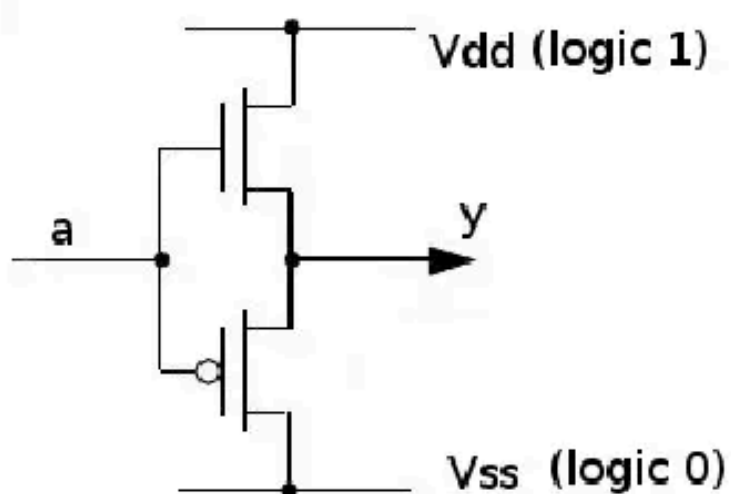
296. What is Inverter? Explain in Cmos Inverter logic?

Ans: It is an electronic device which changes Direct current to Alternating current.



297. What is buffer explanation in CMOS logic?

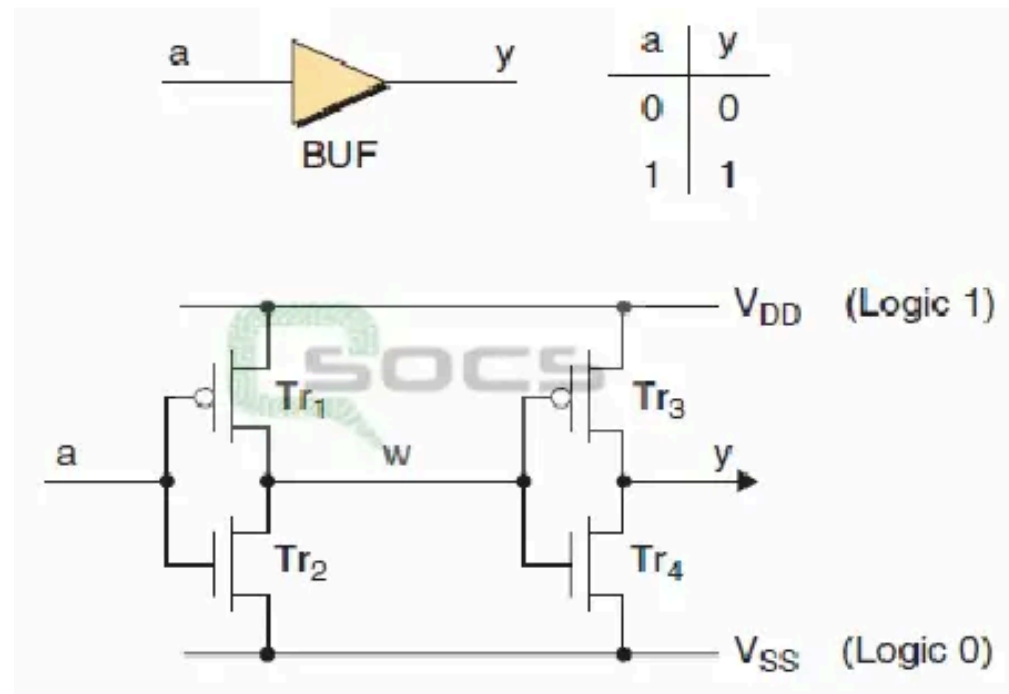
Ans:



By interchanging the positions of the NMOS and PMOS transistors in the NOT circuit can give a Buffer and this technique uses only two transistors.

(OR)

A BUF gate is essentially constructed from two NOT gates connected in series



298. What is the function of AND,NAND,NOT?

Ans: AND: $A*B$

NAND: $AB \text{ Bar}$

NOT : $A \text{ Bar}$

299. What are opens and shots? How u fix?

Ans: Opens: Different metals with same name known as Opens.

How to fix opens:

1. Stretching the metal layers.
2. Metal jogging

Shorts: Same metal with different name known as shorts.

How to fix shorts:

1. Metal shorts:
Metal Jogging (or) change the metal layers.
2. Via shorts:
Changes the vias.

300. What is FV?

Ans: FV (Or) LEC:

It checks the functionality of the netlists

LEC Flow :

Read lib

Read golden netlist

Read revised netlist

Mapping(Names are mapping here)

Comparing

302. What kind of errors u seen in LEC?

Ans: 1. Unmapped names

303. What is Congestion? If congestion is there what happened?

Ans: Available tracks are less than Required tracks known as Congestion.

1. First we need to do set uniform density then also if we observe congestion
2. If Global congestion is there:
 - The congestion which is occurring at every g-cell in the design is called global congestion
 - Due to this global congestion we are facing the improper routing (it will generate opens)
 - This is mainly occurred due to improper placement of macros and bad floor plan
To reduce the global congestion change the core size(increase area) and fixing the proper placement area for std cells.
3. If there are notches congestion there is a possibility to apply hard blockage.
4. If there is more cell density at particular stage we r applying partial blockage and also cell padding.

304. What is CrossTalk? Explain CrossTalk?

Ans: The voltage transition from one net to another net through coupling capacitor known as Crosstalk.

There are two types:

1. Crosstalk Noise
2. Crosstalk Delay

Crosstalk Noise:

If aggressor switches victim is constant known as Crosstalk noise.

If bump is above 50% it goes to metastable state(or) functionality failure.

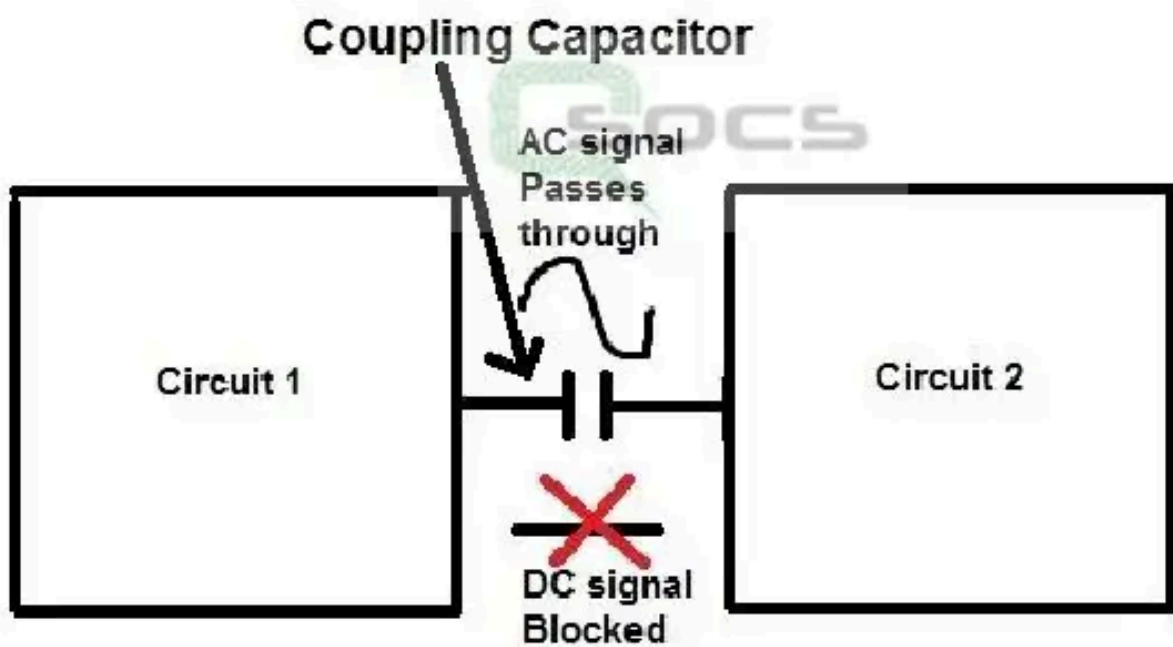
Crosstalk Delay:

If aggressor and victim both in same direction delay decreases.

If aggressor and victim both in opposite direction delay increases.

305. What is Coupling Capacitor?

Ans:



A coupling capacitor is a capacitor which is used to couple or link together only the AC signal from one circuit element to another. The capacitor blocks the DC signal from entering the second element and, thus, only passes the AC signal.

306. Capacitance Formula? Explain in detail?

Ans: It is a passive element that has ability to store the charge in the form of potential difference between plates known as capacitor.

Effect of capacitor is called capacitance.

$$C = \frac{\text{permittivity of dielectric} \times \text{Area of plate overlap in sq.mt}}{\text{Distance between 2 plates}}$$

307. Tell me the manufacture process of metals?

Ans: Metal Manufacturing Processes: **Production**

1. CASTING

Depending on the metal and its purpose, the metal may simply be melted down and molded to shape. This process is known as **casting**. Casting is best for small or intricate parts. **Casting SHOULD NOT be used for products that require high strength, high ductility, or tight tolerances.**

Dies, jewelery, plaques, and machine components all benefit from this simple production process.

2. POWDER PROCESSING

Powder processing treats powdered metals with pressure (pressing) and heat (sintering) to form different shapes. Powdered metallurgy is known for its precision and output quality -- it keeps tight tolerances and often requires no secondary fabrications.

However, it's incredibly costly and generally only used for small, complex parts. **Powder processing is NOT appropriate for high-strength applications.**

3. FORMING

Metal forming takes a raw metal (usually in sheet metal form) and mechanically manipulates it into a desired shape. Unlike casting, metal forming allows for **higher strength, ductility, and workability for additional fabrications.**

Metal Manufacturing Processes: **Fabrication**

1. DEFORMATION

Deformation includes **bending**, rolling, forging, and drawing.

2. MACHINING

Machining refers to any fabrication method that removes a section of the metal. Machining is also known as **material removal processing**. Cutting, shearing, punching, and stamping are all common types of machining fabrication.

When planning for machining in your supply chain, hardening processes should happen AFTER machining processes. Hardened metals have a high shear strength and are more difficult to cut.

3. JOINING

Joining, or assembly, is one of the last steps of the metal manufacturing process. This category includes welding, brazing, bolting, and adhesives. Assembly can be done by machine or by hand.

4. FINISHING

Depending on your material and application, you may also need finishing services. Finishing includes everything from galvanization to powder coating, and can take place throughout the manufacturing process.

308. Why metals are placed in horizontal and vertical fasion?

Ans: At routing stage if we place metals in horizontal and vertical manner it leads to metals lays exactly on tracks.

309. What is CTS? Explain CTS?

Ans: Before CTS we need to check:

1. All cells should be legalized.
2. All power nets are prerouted.
3. All pins should legalized.
4. Congestion, timing should control.

GOALS OF CTS :

1. To minimize the logical DRCs.
2. Balancing the skew.
3. Minimum Insertion Delay.

INPUTS OF CTS:

1. SDC
2. SPEC FILE
3. PLACEMENT DATABASE

WHAT IS CTS ?:

To distribute a clock from Clockport to Clockpin

WHY CTS?:

To minimize skew and insertion delay to build the clock tree.

Here we are generating SPEC file using clock buffers and clock inverters.

SPEC file consists of

1. Buffers list
2. Max skew
3. Min and Max Insertion delay
4. Max trans, Cap, Fanout
5. Inverters list
6. Clock tree leaf pin, exclude pin, stop pin
7. Clock name
8. Clock period

COMMANDS USED IN CTS:

clockdesign
optDesign -postCTS

CHECKS IN CTS:

1. Timing numbers
2. Utilization numbers
3. Congestion
4. All cells should legalize

311. What is CD in shell command?

Ans: change directory

312. how you make a directorie?

Ans: makedirectory(mkdir)

313. Tell me the command for directory with in a directory?

Ans: `mkdir dir1`

